JUBRA

EDUCATIONAL CONSULTS

A Simplified Activity Work Book

PRIMARY FOUR

MATHEMATICS

TERM ONE

NAME: ....

SCHOOL: ....

“ A Journey To Academic Excellency”

PRIMARY FOUR SIMPLIFIED MATHEMATICS TERM ONE

TABLE OF CONTENT

Week 1

Lesson 1

Theme : SETS

Topic : SETS

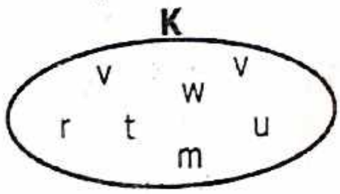
Sub-Topic : Counting number of elements

What is a set:

- A set is a well defined collection of things.
- A set is a collection of well defined elements.
- A member is an object/ item / things /elements that belongs to a set.

Example

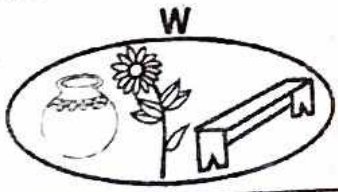
How many elements has set K?



Set K has 7 elements

Example 2.

Find the number of members in set W.

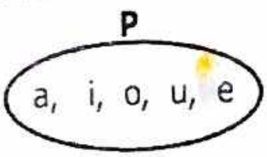


Solution:

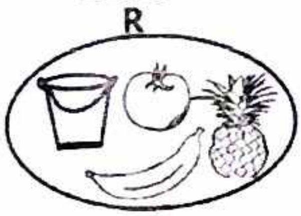
Set W has 3 elements

Activity

Find the number of elements for each set.

1. 

Set **P** has ____________ elements.

2. 

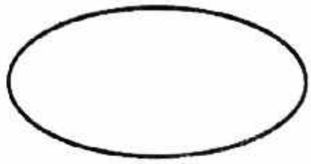
Set **R** has ____________ elements.

3. M: {Samuel, Peter, Betty, Penninah}

How many members has set M?

....

4. K



Set K has ....members.

5. E: {7, 1, 3, 5, 4, 2, 8, 6}

Set E has ....elements.

Identifying sets

Note: To identify sets, we use a capital letter and a common name for all members in that set.

Examples.

Identify the following sets.

1. P

P is a set of 3 balls.

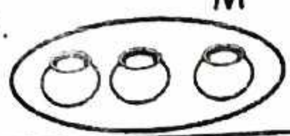
2. Q: {a, e, I, o, u} Q is a set of 5 vowel letters.

Activity.

Identify and count the members in the sets below.

1. B: {0, 1, 2, 3, 4, 5}

2. M

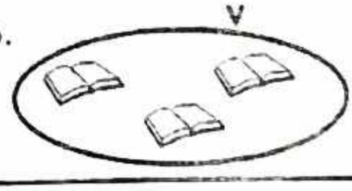


3. N: {buffalo, money, zebra, hyena}

4. D: {a, b, c, d, e}

5. L: {hen, ducky, turkey, pigeon}

6. V

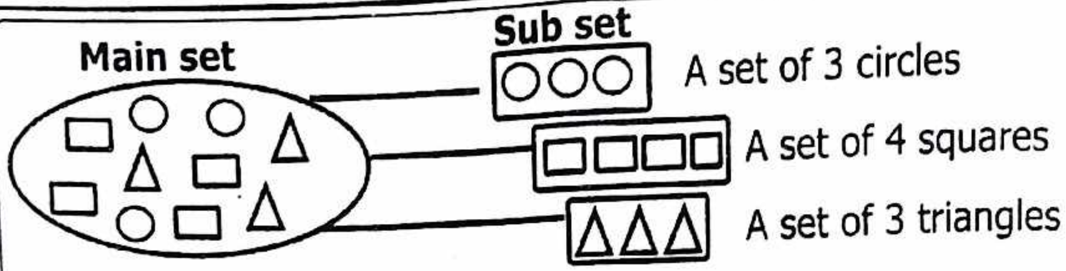


Forming Sets

Sub sets.

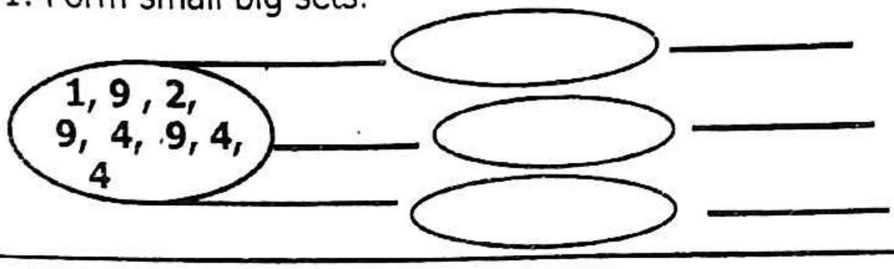
A subset is a small set got from a big set.

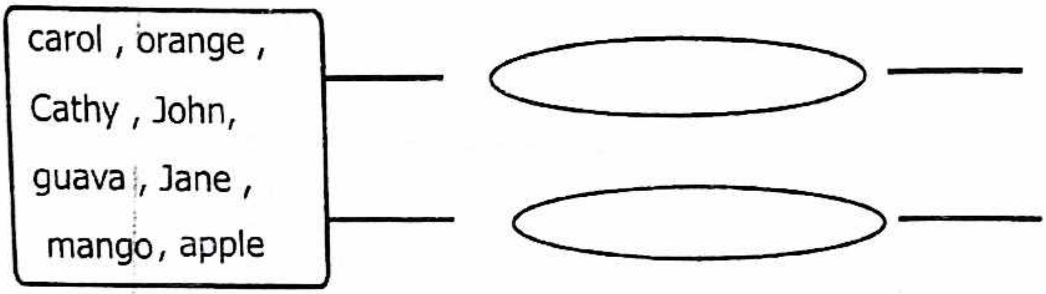
Examples



Activity .

1. Form small big sets.





Week 1

Lesson 2

Content : Naming sets / Types of sets

Types of sets

1. Empty set or Not empty set.

Definition: An empty set is a set with no member or elements.

Symbol for Empty set = $\emptyset$ or { }

Example 1.

Pupils in our class with blue pens.

Solution: Not empty set

Example 2.

A boy in our class who is younger than our teacher.

Solution: Not empty set.

Example 3.

A pupil in our class who come to school by air.

Solution: Empty set.

Activity.

Write **empty set** or **not empty** set.

1. A school whose pupils do not play any game.

____

2. A grand mother of 80 years old. ______________________________
3. A pupil in our class without hands.

____

4. A teacher in our school who is 4 years.

____

5. A pupil in our class with 8 years.

____

6. A goat which is eaten as food. ______________________________

Week 1

Lesson 3

Content : Equivalent sets and non equivalent set.

Equivalent set are set with the same number of members but they are not the same. $\longleftrightarrow$ or $\equiv$

Non equivalent set is a set which do not have the same number of elements

Symbol $\not\longleftrightarrow$

Example 1.

Set M ={1, 2, 3, 4, 5} and Set N ={ a, e, i, o, u }

Set M has 5 members and set N or **set M** $\equiv$ **to set N.**

Set M $\longleftrightarrow$ **to set N.**

Example 2.

Given that set P ={ star, rectangle, circle} and set Q ={m , e , a , t }

Solution

Set P has 3 elements.

Set Q has 4 elements.

$\therefore$ **Set P and set Q are non equivalent sets.**

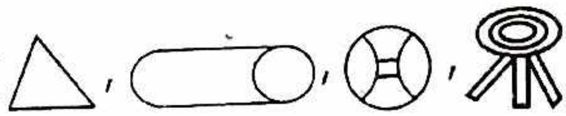
Set P $\not\leftrightarrow$ to set Q. or Set P $\not\equiv$ to set Q.

Activity.

Use equivalent sets or non equivalent set.
($\equiv$ or $\leftrightarrow$ / $\not\equiv$ or $\not\leftrightarrow$ to complete the sentences.

1. A ={ car, cat , kettle) and B={ chair, bench, pen}
   Set **A** has __________members and
   set **B** has __________members.
   Set A and Set B are ______________________________.

2. Given that set D ={a, e, i, o, u} and set E= { 1, 2, 3, 4, 5}
   Set D has __________elements and
   Set E has ____________elements.
   Set D ______________Set E.

3. If set W ={  } and X={  }.

   Set W has __________and set X has ______________.
   Set W has ______________set X.

Week 1

Lesson 4

Content : Even and odd sets .

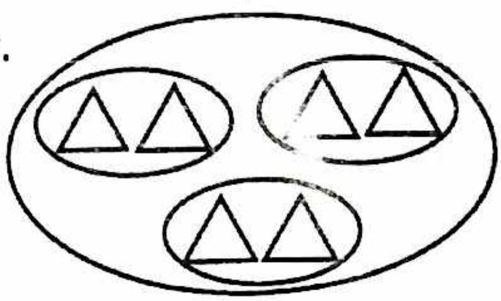
Note: Even sets are sets whose members can all be paired.

- Odd sets are sets whose elements cannot be all paired i.e they give a reminder when their elements are paired.
- An empty set is an even set.

Example:

Given set Y below.

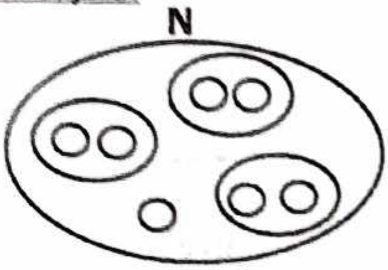
Y



Set Y has 6 elements

Members of set Y have all been repaired , therefore it is an even set.

Example:

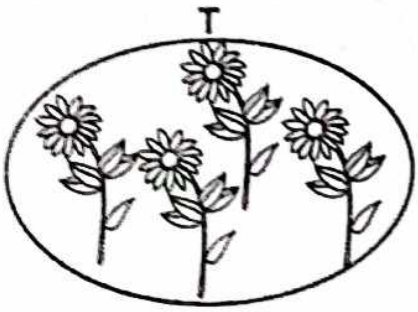


Set **N** has 7 elements.
Members of set N have not been paired.
Therefore it is an odd set.

Activity.

Use the diagram below to answer questions below.

1.



a) How many elements has set T?

....

b) How many pairs of flowers are in set T?

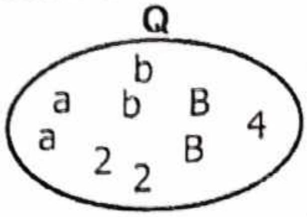
....

c) How many members are not paired?

....

d) Set T is an.........................set.

2.



a) How many elements has set Q?

....

b) How many pairs have been formed from set Q?

....

c) How many elements were not paired in set Q?

....

Week 1

Lesson 5

Content : Intersection set

Note: Intersection are sets with common members.

- Symbol for intersection set "∩"
- Sets with common members are also called intersecting sets

Example 1.

Set P={a, b, c, d, e} and
Q={a, e, i, o, u}

1. Find PnQ

Solution

Set P={~~a~~, b, c, d, ~~e~~} and
Q={~~a~~, ~~e~~, i, o, u}

PnQ ={ a, e}

(ii) Find n(PnQ)

Solution

PnQ = {a, e}

;. n(PnQ) = 2 elements

Note: i) Set **P** and **Q** are intersection sets because they have common members.

Example 2.

Given that set F={4, 5, 6, 7} and G ={x, y, z, m}

i) Find FnG

Solution .

FnG = { }

ii) Find n(FnQ)

Soln.

FnG = { }

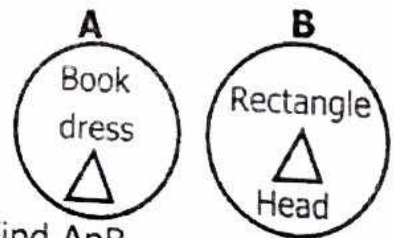
∩(∩Q) = 0 elements

F n G

Set F and G are non -intersection sets because they don't have any common members.

Activity

1.

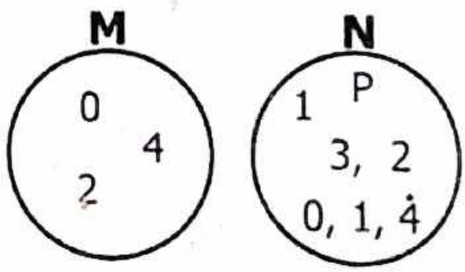


i) Find AnB

ii) Find n(AnB)

iii) Set A and B are .... sets.

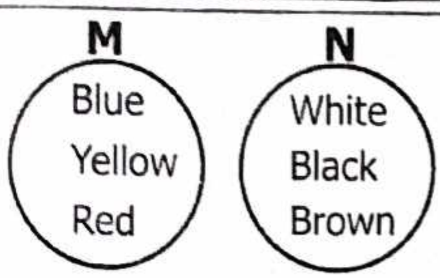
2.



i) Find MnN

ii) Find n(MnN)

iii) Set M and set N are ....sets.

3.


I) Find set EnF

ii) Find n(EnF)

iii) Set E and set F are

....sets

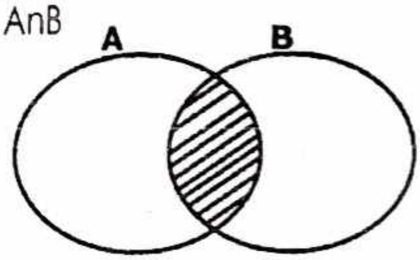
Week 1

Lesson 6

Content : Drawing venn diagrams and shading the intersection region

Example 1:

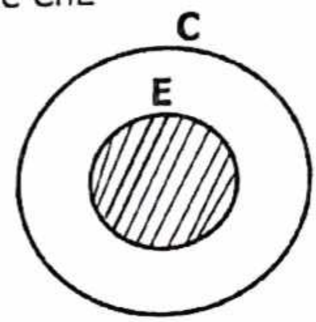
Shade AnB



AnB is shaded

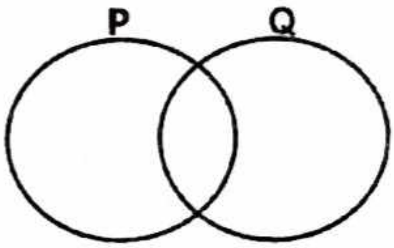
Example 2.

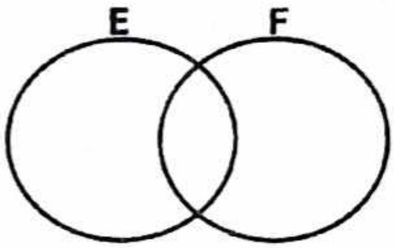
Shade CnE

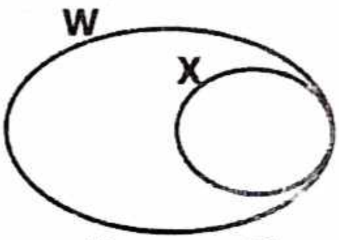


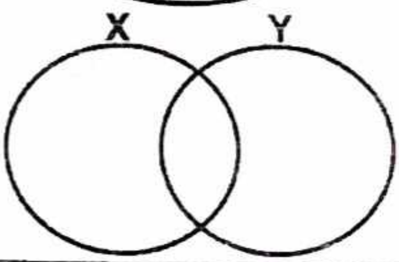
Activity.

In the venn diagrams below shade the intersection sets.

1.


2.


3.


4.


Week 1

Lesson 7

Content : Listing elements of intersection.

Example 1:

Given that set K = {0, 1, 2, 3, 4, 5}

and set L={1, 3, 5, 7}

1. List the elements of KnL.

Solution

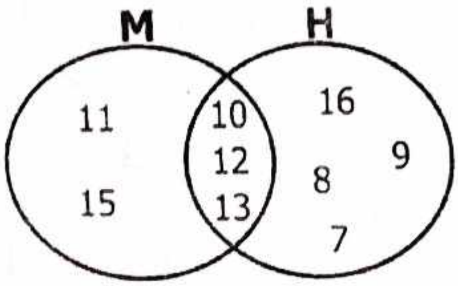
K ={0, 1, 2, 3, 4, 5}

L = {1, 3, 5, 7 }

KnL = { 1, 3, 5 }

Example 2.

In the venn diagram below, list the elements of NnM



solution

MnN = {11, 12, 13}

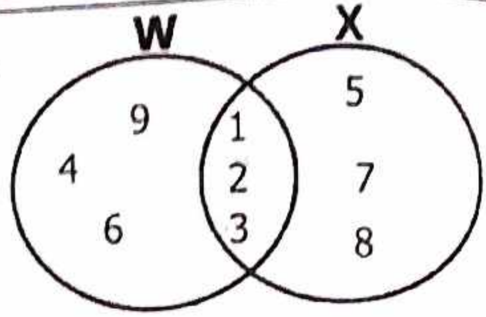
Activity.

List the elements of intersection in the following sets.

1. P= {  } and

Q= {  }

2.



3. A ={ 1, 2, 3, 4, 5, 6, 7, 8 }
B={ 0, 2, 4, 6, 8, 10}

4. S ={ Sam, John, sister, man}
R={ sister, Bidi, Joan, Sarah}

Week 1

Lesson 7

Content : Finding number of elements of intersection.

Example 1:

If set A ={ a, b, c, d, e, f, g} and set B ={ a, e, i, o, u}

1. Find n(AnB)

Solution

A ={a, b, c, d, e, f, g}

B = {a, e, i, o, u}

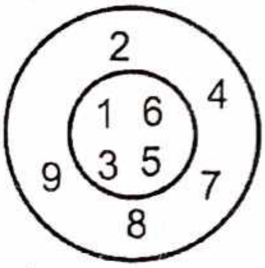
(AnB)={ a, e }

n(AnB) = 2 elements

Example 2:

In the diagram below.

Find n(PnQ)



Solution:

PnQ = { 1, 3, 5, 6 }

.. n(PnQ) = 4 elements

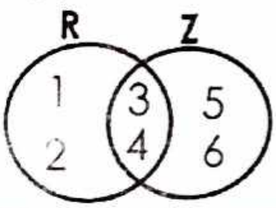
Activity.

1. Set W = { m, a, n} and
   Set N ={w, o, m, a, n}
   Find n(MnN)

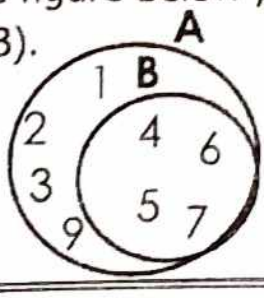
2. Set T = { y, e, l,l,o,w} and
   Set G ={R, e, d}
   Find n(TnG)

3. Set y = { p, o,t} and
   Set S ={S, p, o, r, t s}
   Find n(YnS)

4. Find (i) RnZ
   (ii) Find n(RnZ)



5. In the figure below , workout n(AnB).



Week 2

Lesson 1

Content : Union sets

Definition:

Union sets is a collection of all the members in the given sets.

Symbol = ∪

Listing members of the Union set.

You have to first identify the common elements to avoid repeation of recording them.

Example 1:

If set W ={J, A, B, S } and
set Y ={ M, J, S, W, }

List the members of WUY.

Solution:

W ={J, A, B, S}

Y = {M, J, S, W}

WUY ={J, A, B, S, M, W}

Example 2:

If set P ={ a, e, i, o, u} and
set Q ={ a, b, c, d, e }

Use the sets above to find PUQ

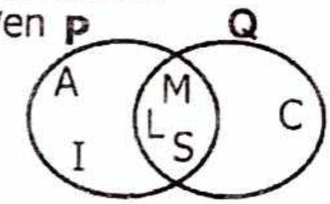
P={a e i, o, u}

Q= {a b, c, d, e}

PUQ = {a, e, i, o, u, b, c, d}

Activity.

1. Given



Find PUQ

2. E={2, 4, 5, 6, 7, 8} and
F={3, 4, 6, 8, 9}
Work out EUF.

3. **P** = {x, y, z, t}
F={a, e, i, o, u}

P u F = {__________________}

4. Given N={p, o, r, t}
and M ={p, a, r, t}
Find NUM

5. If X={  }

and Y ={  }
Find XUY

Week 2
Lesson 2
Conten : Finding number of elements of union sets.

Example 1:
If E = {w, o, m, a, n} and
H={s, p, o, r, t, s, m, a, n}
Find n(EUH)

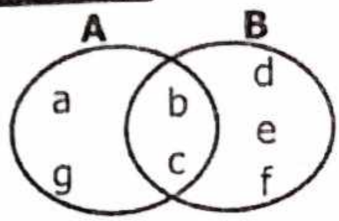
Solution:
E = {w, ~~o~~, ~~m~~, ~~a~~, ~~n~~} H={s, p, ~~o~~, r, t, s, ~~m~~, ~~a~~, ~~n~~}
EUH ={w, o, m, a, n, s, p, r, t}
n(EUH) = 9 elements

Example 2:
If Q ={1, 2, 3, 4, 5} and
R={2, 3, 5, 11}. Find n(QUR)

Solution.
Q ={1, 2, 3, 4, 5}
R={2, 3, 5, 7, 11}
(QUR) = {1,2,3,4,5,7,11}

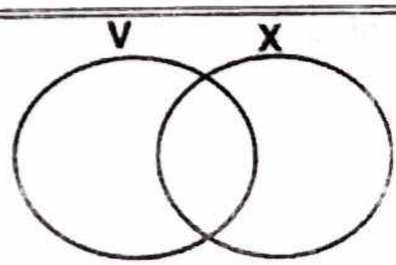
n(QUR) = 7 elements

Activity.

1.


Workout n(AUB)

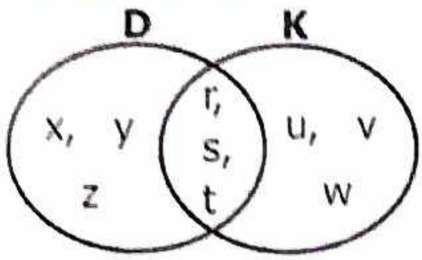
2. P={0, 1, 2, 3, 4, 5}
Q ={2, 4, 6, 8}
Find n(PUQ)

3.


Find n(VUX)

4. X={book, box, board} and
Y= {pen, book, pencil}
Find n(XUY)

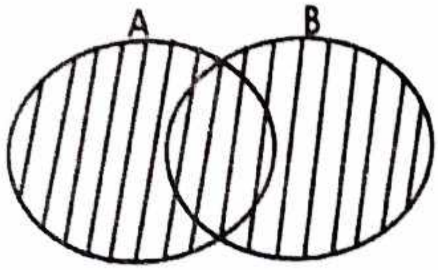
5.



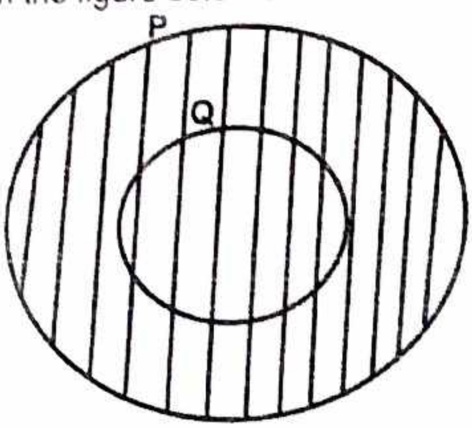
Find n(DUK)

Week 2
Lesson 3
Content : Drawing and shading Union of sets.

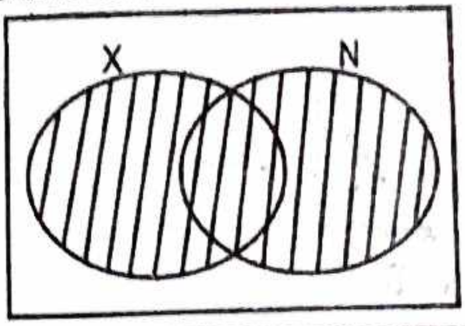
Example 1:
Shade AUB



Example 2 :
In the figure below shade PUQ.



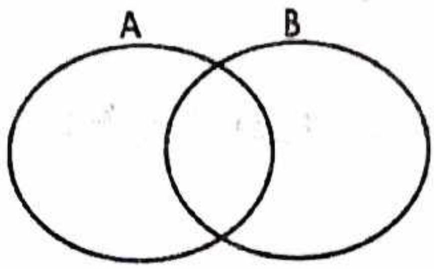
Example 3 :
Shade XUN



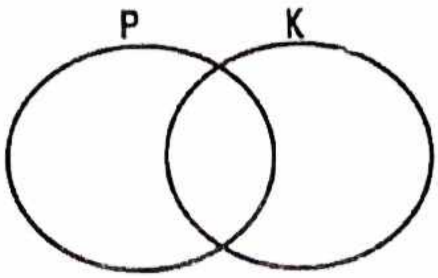
Activity.

In the following venn diagrams below shade the Union of the sets given below.

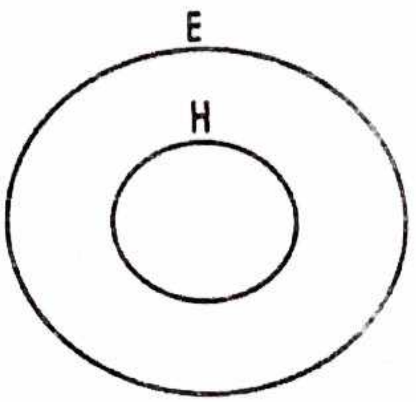
1.



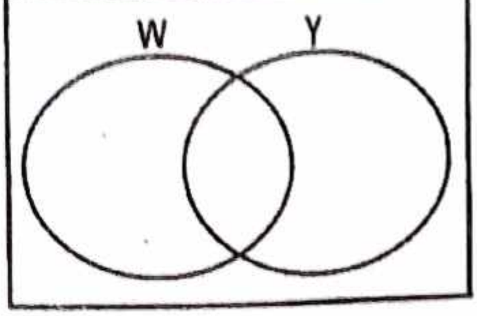
2.



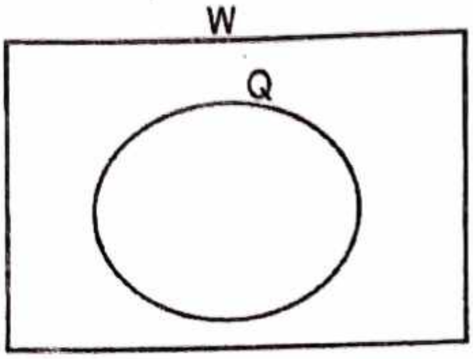
3.



4.



5.



Week 2
Lesson 4
Content : Difference of sets.
Definition:
Difference of sets are members of a set that exist in only one set.

Example
A-B means members of set A only
or
A-B means elements of set A which are not found in set B.

Example1.
Given A ={1, 2, 3, 4, 5, 6, 7} and
B={ 1, 3, 5, , 9, 11, 13}
Solution :
Using the above sets, find i) A - B
A - B = {2, 4, 6}

ii) Find n(A-B)
n(A-B)={2, 4, 6}
n(A-B) = 3 elements

iii) Find B only
Solution.
B only = {9, 11, 13}

iv) Find n(B-A)
Solution.
B-A = {9, 11, 13}
n(B-A) = 3 elements

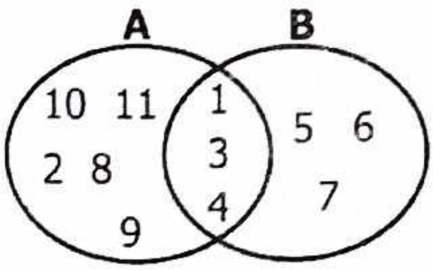
Activity.
1. X ={ s, p, o, r, t} and
Y ={ t, e, r, m}
i) Find X-Y

ii) Find Y-X

iii) Find n(X-Y)

iv) Find n(Y-X)

2. Use the venn diagram below to answer questions that follow.



Find (i) A-B

(ii) B only

iii) n(A-B)

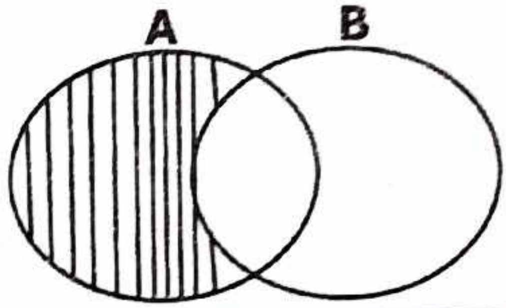
iv) n(B-A)

Week 2
Lesson 5

Content : Drawing and shading of sets.

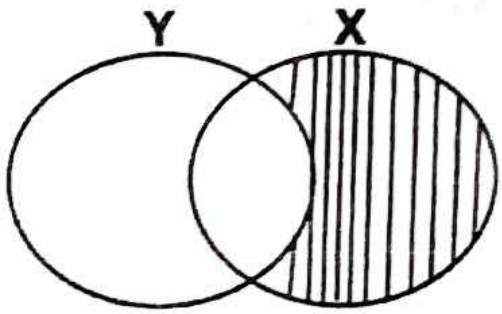
Example 1.

Shade A-B in the diagram below.



Example 2.

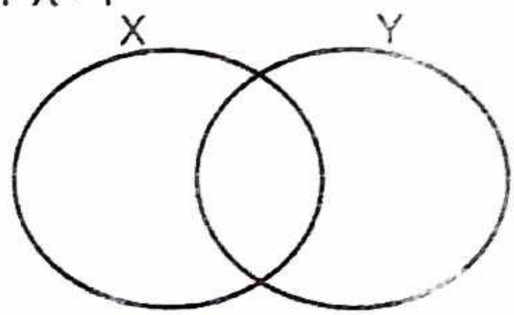
Shade X-Y in the venn diagram below.



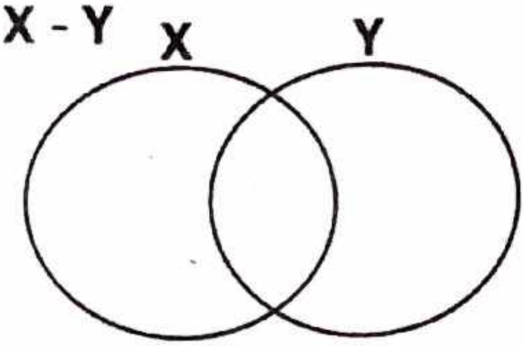
Activity.

Use the venn diagrams below and shade the following .

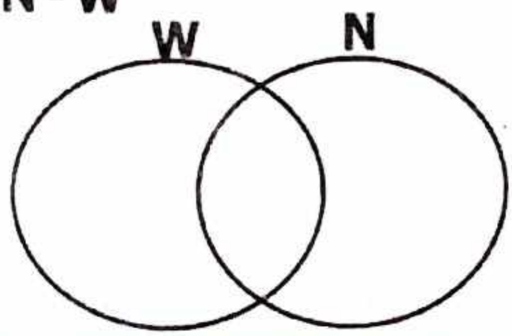
1. X - Y



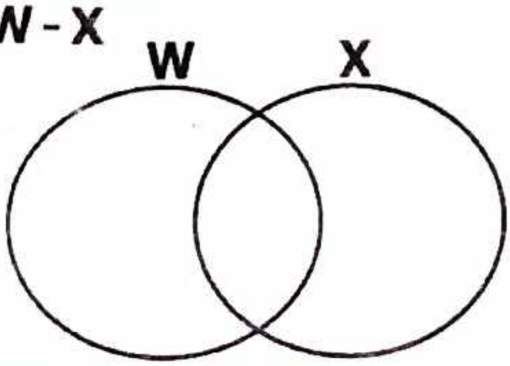
2. **X - Y**



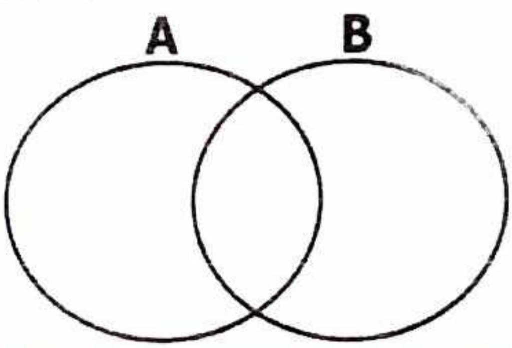
3. **N - W**



4. **W - X**



5. Shade B - A



Week 2

Lesson 5

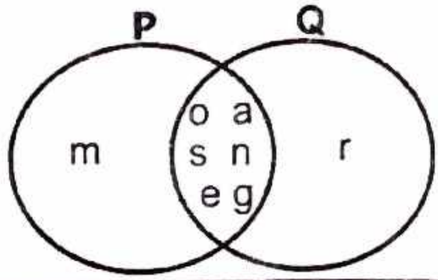
Content : Representing element on venn diagram

Example.

Given P={m, a, n, g, o, e, s}

Q={o, r, a, n, g, e, s}

a) Represent the above information on venn diagram below.



b) Find PnQ

Solution

PnQ ={o, a, n, s, e, g}

c) Find Q-P

Solution

Q-P ={r}

d) Find n(PUQ)

Solution

PUQ ={m, o, a, n, s, e, g, r }

n(PUQ) = 8 elements

Activity

Given A={y, a, r, m, s}

B={p, a, m, e, a, p}

a) Represent the above information on a venn diagram.

b) Find ; i) AnB

ii) n(AnB)

iii) B - A

IV) n(A-B)

2. X={

}

Y

}

i) Represent the information on the venn diagram.

ii) Find XnY.

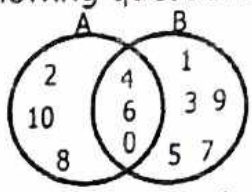
iii) What is X - Y?

iv) Find n(Y - X)

Interpreting information on the venn diagram.

Examples.

Study the venn diagram below and use it to answer the following questions.



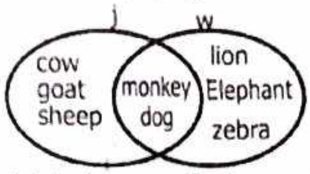
a) List down the elements of set A.
Set A {0,2,4,6,8,10}

b) How many members are in set B.
Set B= {4,6,0,1,3,5,7,9}
= 8 members.

c) Find;
i) AUB = {0,10,8,2,4,1,3,5,7,9}
ii) A∩B = {2, 4, 6}

Activity.

1. Study the diagram below and use it to answer the questions that follow.



a) List down all elements of set K.

b) How many members are in set J?

c) Find: i) $J \cap K$

ii) $J \cup K$

d) How many members are in set JUK?

1. X = { }

2. P = {1, 2, 3}

3. A ={ a, b}

4. B ={ c }

5. { a, b, c, d}

Week 2
Lesson 6
Content : Subsets

Note: - A subset is a set of members got from a given set.
- An empty set is a subset of any set.
- A set itself is a subset.
- Another set is a subset itself.

Symbol = $\subset$

Example 1.

List all the possible subsets of set Q={1, 2}

Soln.

{1,2} ={ } , {1 } , {2}, {1, 2}

ACTIVITY

List all the possible subsets of the following sets.

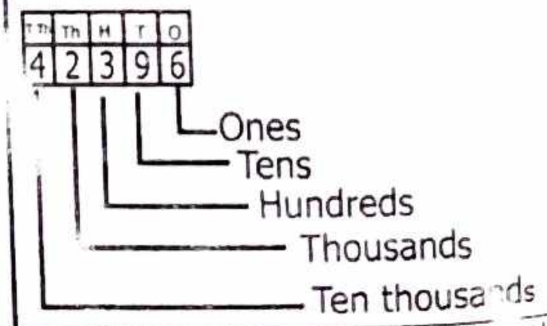
Week 2
Lesson 7
Content : Whole numbers

Writing place values of digits in given numbers

Example

Write the place value of each digit in 42396

Solution



Example 2.

Write the place value of 2 in 246369

HTh	Tth	Th	H	T	O
2	4	6	3	6	9

Ones
Tens
Hundreds
Thousands
Ten thousands
Hundred thousands

The place value of 2 is hundred thousands.

ACTIVITY

1. Write the place value of each digit in the following numbers.

a) 423

b) 6940

c) 32324

d) 21000

e) 1000

f) 14

g) 2

2. Write the place value of the underlined digits.

a) 4 $\underline{6}$ 3 9

....

b) 2 9 3 $\underline{1}$

....

c) $\underline{4}$ 2 2 98

....

Week 3
Lesson 1
Content : Writing values of digits in given numbers.

Example 1:
Write the value of each digit in

H	T	O
4	2	3

Ones = 3 x 10 =3
Tens = 2 x 10 = 20
Hundreds=4x100 =400

Example 2:
Write the value of 7 in 6793.

Th H T O
6 7 9 3.

Solution:
Value = Digit x place value
= 7 X 100
= 700

Activity

1. Write the value of each of digit in the following.

a) 42

b) 161

c) 427

d) 879

e) 4693

f) 63694

2. Write the value of the underlined digits.

a) 639 = ____________

b) 3694 = ____________

c) 2361 = ____________

d) 66397 = ____________

Week 3
Lesson 1
Content : Writing values of digits in given numbers.

Example 1:
What is the value of 6 tens?
Soln.
Value = Digit x place value
= 6 x tens
= 6 X 10
= 60

Example 1:
What is the value of 16 hundreds?
Soln.
Value = Digit x place value
= 16 x hundreds
= 16 x 100
= 1600

ACTIVITY

Find the value of the following.

1. 4 tens

2. 142 tens

3. 7 hundreds

4. 169 thousands

5. 100 tens

6. 31 tens

7. 14 hundreds

Finding sum and difference of values of digits .

Examples.

Find the sum of the values of 4 and 9 in the number 4392?

Th	H	T	O
4	3	9	2

9 X 10=90

4x1000=4000

$$\begin{array}{r} 4000 \\ +\ 90 \\ \hline 4090 \\ \hline \end{array}$$

2) What is the difference between the value of 4 and 7 in the number 3547?

Th	H	T	O
3	5	4	7

7x1 = 7

4x10=40

$$\begin{array}{r} 40 \\ -\ 7 \\ \hline 33 \\ \hline \end{array}$$

Activity

1. What is the sum of the value of 9 and 4 in the number a) 4,392

2. What is the sum of values of 8 and 2 is the number 3284?

3. Find the difference between the value of 6 and in 7 the number 67.

4. Work out the difference between the value of 7 and 4 in the number 17304.

8. 8 ones

Week 3

Lesson 3

Content : Writing figures in words.

Example 1:

Write 617 in words.

Sln.

Units		
H	T	O
6	1	7

617 = six hundred seventeen

Example 2:

Write 594 in words.

Soln.

Units		
H	T	O
5	9	4

594 = Five hundred ninety four.

or

Five hundred and ninety four.

Example 3:

Write 6937 in words.

Soln.

Thousands			Units		
H	T	O	H	T	O
		6	9	3	7

6,937 = **six thousand, nine hundred thirty seven**

Activity.

Write the following in words.

1. 146

2. 213

3. 193

4. 114

5. 18211

6. 92412

7. 92412

8. 1224

Week 3

Lesson 4

Content : Writing number word in figures.

Example 1:

Write twelve thousand , eight hundred sixty four.

Soln.

Twelve thousand = 12000
Eight hundred = 800
forty one = 41
= 12841

Example 2:

Write thirty six thousand, nine hundred sixty four.

Solution:

Thirty six thousand = 36000
nine hundred sixty four = 964
= 36964

Activity

Write the following in Hindu-Arabic (figures)

1. Seventy one thousand, two hundred forty four.

2. Four hundred ninety four

3. Five thousand, one hundred one.

4. One thousand six hundred.

5. Twenty six thousand, eight hundred forty one.

6. Thirty one thousand, one hundred one

3. 27428

4. 7862

Week 3

Lesson 5

Content : Writing numbers in expanded form using place values.

Example 1:

Expand 639 using place value.

Soln.

H T O

6 3 9 = 600 + 30 + 9

= (6x100)+(3x10)+(9x1)

5. 993

Example 2:

Expand 3926 using place value.

Soln.

Th H T O

3 9 6 2 = 3000 + 900 + 60 + 2

=(3x1000)+(9x100)+(6x10)+(2x1)

6. 612

Activity.

Expand the following numbers using place values.

1. 341

7. 14

8. 235

2. 142

Week 3
Lesson 6
Content : Writing numbers in expanded form using values.

Example 1:
Expand 639 using value.
Solution

H T O
6 3 9 = (6x100)+(3x10)+(9x1)
= 600 + 30 + 9

Example 2:
Expand 3975 using value.
Solution
3975=(3x100)+(9x100)+(7x10)+(5x1)
= 3000 + 900 + 70 + 5

Activity

Expand the following using values.

1. 868

2. 426

3. 46

4. 3621

5. 9246

6. 786

7. 7040

8. 8435

Week 3
Lesson 7
Content : Writing expanded numbers in short form

Example 1:
What number has been expanded to give;
(3x1000) + (4x100) + (6x10) + (8x1)

Solution
=(3x1000) + (4x100) + (6x10) + (8x1)
= 3000 + 400 + 60 + 8
= 3000
400
60
8
= 3468

Example 2:
Find the number which has been expanded to give;
9000+400+60+1

Solution
= 9000+400+60+1
9000
400
60
+ 1
= 9461

Activity.
What number has been expanded to give;
1. 3000 + 600 + 80 + 7

2. (3x100) + (6x10) + (1x1)

3. 400 + 30 + 2

4. 500 + 70 + 2

5. (6x100) + (9x100) + (4x1)

6. 7600 + 300 + 40 + 2

Week 4
Lesson 1
Content : Writing place values of whole and decimals.

Example 1:
Write the place value of each digit in 4.32.

Solution
4.32
2 – Hundredths
3 – Tenths
4 – Ones

Example 2:
Write the place value of each digit in 10.639. **solution.**
110. 639
9 – Thousandths
3 – Hundredths
6 – Tenths
0 – Ones
1 – Tens
1 – Hundreds

Activity.

Write the place value of each of the following.

1. 90.2

2. 0.42

3. 6.3

4. 118.693

5. 6.3

6. 2.0064

Week 4

Lesson 2

Content : Writing decimals in words.

Example 1:

Write

4.3 = four point three

or 4.3 = four and three tenths.

Example 2

Write 0.6 in words.

Soln.

0.6 = Zero point six or six tenths

Example 3

Write 4.03 in words.

Soln.

4.03 = four point zero three

or Four and three hundredths

Activity.

Write the following in words.

1. 0.6

2. 3.09

3. 143.6

4. 24.42

5. 6.31

6. 0.01

6. 0.01

7. 0.42

8. 0.34

Week 4

Lesson 3

Content : Rounding off to the nearest tens.

Example 1:

Round off 432 to the nearest tens.

Soln.

Note : When rounding off numbers learners must note the following:

1. Identify the digit in the position of the required place value.
2. Identify the digit on the left of the digit in the position of the required place value.
3. Compare the digit on the left if it is less than 5 i.e 0, 1, 2, 3, 4. We add 0 to the digit in the position of the required place value.

If the digit is greater that 5 or equal to 5 i.e 5, 6, 7, 8, 9, we add 1 to the digit in the required place value.

4. All digits behind the digit in required place value are replaced by zeros.

Solution

2 was replaced with zero

H T O

4 $\boxed{3}$ 2 = 430

= 430

+ 0

430

$432 \simeq 430$

Example

Round 469 to the nearest tens

Solution

9 was replaced with zero

H T O

4 $\boxed{6}$ 9 = 460

=460

+1

470

$469 \simeq 470$

H T O

4 6 9

RPV

Activity.

Round off the following to the nearest tens.

1. 463

2. 698

3. 460

4. 6930

5. 1498

6. 63

7. 99

8. 4938

Week 4
Lesson 4
Content : Round off numbers to the nearest hundreds.

Example 1:
Round off 1496 to the nearest hundreds.

Th H T O
1 $\boxed{4}$ 96 = 1400
= +1
1500

. 1496 ≃ 1500

Example 2:

Th H T O
6 $\boxed{9}$ 36 = 6936
= + 0
6900
6900

Activity

Round off the following to the nearest hundreds.

1. 469

2. 146

Rounding off using a number line.
Examples.

Round off 47 to the nearest tens using a number lies.

40 41 42 43 44 45 46 47 48 49 50

47 is nearer to 50 than 40.
47 ≏ 50

2) Round off 61 to the nearest tens using a number line

60 61 63 63 64 65 66 67 68 69 70

61 is nearer to 60 than 70.
61 ≏ 60

Activity

1) Round off 43 to the nearest tens using a number line.

2) Round off 26 to the nearest tens using a number line.

3) Round off 18 to the nearest tens using a number line.

3. 2693

4. 1950

5. 639

6. 6954

7. 819

8. 889

Week 4

Lesson 5

Content : Rounding off numbers to the nearest thousands.

Example 1:

Round 1496 to the nearest thousands.

Solutions

Th H T O

$\boxed{1}496 = 1000$

$= +0$

$\underline{1000}$

. $1496 \simeq 1000$

Example 2:

Round off 4849 to the nearest thousands.

Solutions

Th H T O

$4\boxed{8}49 = 4000$

$= +1$

$\underline{5000}$

. $4849 \simeq 5000$

Activity.

Round off the following to the nearest thousands.

1. 4373

2. 14296

3. 3693

4. 4265

5. 3210

6. 1400

7. 3840

8. 19490

Week 4

Lesson 6

Topic: Whole Numbers

Sub-Topic: Roman Numerals

Content: Basic Roman Numeral

General Roman Numerals

Hindu-Arabic	1	5	10	50	100	500	1000
Roman Numerals	I	V	X	L	C	D	M

Roman Numerals obtained by repeated addition.

Hindu-Arabic	2	3	6	7	8	20
Roman Numerals	II	III	VI	VII	VIII	XX

Hindu-Arabic	30	60	70	80	200	300
Roman Numerals	XXX	LX	LXX	LXXX	CC	CCC

Roman Numerals obtained by subtraction.

Hindu-Arabic	4	9	40	90	400	900
Roman Numerals	IV	IX	XL	XC	CD	CM

5-1 10-1

Note: :Roman Numerals are only and only expressed in capital letters

Expressing Hindu -Arabic Numerals into Roman Numerals

Example 1.

Express 38 into Roman Numerals.

Solution.

38 = 30 + 8

= XXX + VIII

= XXXVIII

Example 2.

Write 146 into Roman Numerals.

Solution.

146 = 100 + 40 + 6
= C + XL + VI
= CXLVI

Activity.

Express the following into Roman numerals.

1. 19
2. 34
3. 58
4. 99
5. 145
6. 128
7. 199
8. 569
9. 1699
10. 156

Week 4

Lesson 7

Content: Expressing Roman Numerals into Hindu- Arabic Numerals.

Example 1.

Express XLIV in Hindu-Arabic Numerals

Solution.

XLIX = XL + IX
= 40 + 9
= 49

Example 2.

Write CXCIX in Hindu-Arabic Numerals.

Solution.

CXCIX= C + XC + IX
= 100 + 90 + 9
= 199

Activity.

Write the following Roman Numerals into Hindu-Arabic numerals

1. CLIII

2. XIX

3. CV

4. LXXXVIII

5. XXXVIII

6. XXVI

7. XLIII

8. XXXIV

9. CXLIV

Week 5

Lesson 1

Content: Evaluating facts on Roman Numerals.

Example 1.

Express is 28 years old. Express his age in Roman numerals.

Solution

29 = 20 + 8

= XX + VIII

= XXVIII years

Example 2.

There are XLIX pupils in our class. Write this number in Hindu-Arabic numerals.

Solution

XLIX = XL + IX

= 40 + 9

= 49 pupils.

Activity.

1. Kaketo is 38 years old. What is the age in Roman numerals.

2. Shafik's vehicle has been driven for 24 months. Write these months in Roman numerals.

3. It is XV minutes past IV clock. Write the time in Hindu-Arabic numerals.

4 . Sam carried a jerrycan of XL litres and John carried a jerrycan of XV litres. Find the total in Hindu Arabic numerals.

5. If it is 45km from our school to the home. What is this distance in Roman numerals.

Week 5

Lesson 2

Content: Addition of Roman Numerals.

Example 1.

Add IX + V

Solution

= IX + V
= 9 + 5
= 14
= 10 + 4
= X + IV
= XIV

Example 1.

Add XX + VII

Solution

= XX + VII
= 20 + 7
= 27
= 20 + 7
= XX + VII
= XXVII

Example 3.

Add XLVII + XXV

Solution

= XLVII + XXV
= 47 + 25
= 72
= 70 + 2
= LXX + II
= LXXII

Activity.

Add the following

1. X + VII

2. XXXVI + XXII

3. IX + V

4. XL + XIX

5. XCII + XL

6. XLIX + XII

7. XL + L

8. XVIII + XL

Week 5

Lesson 3

Content: Subtraction of Roman numerals.

Example 1.

Subtract: XXXVI - XXVIII

Solution

= XXXVI - XXIII
= 36 - 23
= 13
= 10 + 3
= X + III
= XXIII

Example 2.

Workout: XL - IX

Solution

= XL - IX
= 40 - 9
= 31
= 30 + 1
= XXX + I
= XXXI

Activity.

Subtract the following.

1. XXX - XXIX

2. XCIII - LXXIII

3. XX - IX

4. Subtract XII from XXIX

5. XLIX - XXIV

6. Takeaway XIX from XX.

Topic: Operation on whole numbers.

Week 5

Lesson 4

Content: Addition of whole numbers

Addition of whole numbers without regrouping.

Examples 1.

Add 463 + 534

Soln.

$$
\begin{array}{rl}
463 & = 400 + 60 + 3 \\
+\ 534 & = 500 + 30 + 4 \\
\hline
 & = 900 + 90 + 7 \\
\hline
 & = 900 \\
 & = 90 \\
 & = +\ 7 \\
\hline
 & = 997 \\
\hline
\end{array}
$$

Method 2.

$$
\begin{array}{rl}
\text{H T O} & \\
4\ 6\ 3 & \text{- Add ones } 3 + 4 = 7 \\
+\ 5\ 3\ 4 & \text{- Add tens } 6 + 3 = 9 \\
\hline
9\ 9\ 7 & \text{- Add hundreds } 5+4=9 \\
\hline
\end{array}
$$

Examples 2.

Add:

$$
\begin{array}{r}
6\ 3\ 4 \\
+\ 1\ 4\ 3 \\
\hline
\\
\hline
\end{array}
$$

Soln.

$$
\begin{array}{rl}
634 & = 600 + 30 + 4 \\
+\ 143 & = 100 + 40 + 3 \\
\hline
 & = 700 + 70 + 7 \\
\hline
 & = 700 \\
 & = 70 \\
 & = +\ 7 \\
\hline
 & = 777 \\
\hline
\end{array}
$$

Method 2.

```
 H T O
 6 3 4 - Add ones 4 + 3 = 7
+1 4 3 - Add tens 3 + 4 = 7
 7 7 7 - Add hundreds 6+1=7
```

ACTIVITY

Add the following:

1.
```
  2 4 6
+ 3 6 9
```

2. 1 4 6 + 3 5 2

3. 1463 + 421

4.
```
  1 1 4
+ 6 6 3
```

5. 7 + 2

6. 17 + 81

Week 5

Lesson 5

Content: Addition of whole numbers with regrouping.

Example 1. Method 1

```
 H T O
 4 6 9 - Add ones 9 + 9 = 18
  +6 9 - Add tens 1+6+6 =13
 5 3 8 - Add hundreds 1+4=5
```

Method 2

Soln.

```
 4 6 9 = 400 + 60 + 9
+  6 9 = 000 + 60 + 9
         = 400 +120 +18
         =400
         = 120
         =  18
         = 538
```

Example 2. Method 1.

Add 14963 + 6893

Soln. Method 1

```
 14963=10000+4000+900+60+3
+6893=00000+6000+800+90+3
     =10000+10000+1700+150+6
     =10,000
     =10,000
     =  1700
     =   150
     =     6
     = 21856
```

Method 2.

Hth Th H T O
 1 4 9 6 3 - Add ones 3+3 =6
+ 6 8 9 3 - Add tens 6+9 =15
 2 1 8 5 6 - Add hundreds=1+9+8 =18
- Add thousand=1+4+6 =11
- Add ten thousand 1+1= = 2

Activity.

Add the following

1. 4 6 3 9
+ 8 8 9 9

2. 463 + 889

3. 6969 + 3693

4. 7546 + 8632

5. 184 + 998

6. 99 + 87

7. 19 + 99

8. 185 + 95

Week 5

Lesson 7

Content: Evaluating facts on addition of whole numbers.

Example 1.

A lorry carried 349kg of maize and a truck carried 578kg of maize. How many kilograms of maize were carried altogether?

Solution:

A lorry carried = 349kg
A truck carried = 578kg
= 927kg

Example 2.

Kato bought 5kg of rice shs.2500 and he also bought carry powder at shs. 5400. How much did he used altogether?

Solution:

	Rice	= shs. 2500
+	Carry powder	= shs. 5400
		= sh. 7900

Activity

1. A school has 440boys and 839girls. How many pupils are there in the school?

2. Find the sum between 146 and 249.

3. A farmer planted 1264 banana suckers this year and he is hopping next year to plant 4639 banana. How many banana plantations will he have altogether

4. At Kamukamu shop they sold Bukedde news at sh. 2500 and Orumuri news at shs.1500. How much did they get altogether if they sold 100 copies of Bukedde and 50 copies of Orumuri.

Week 6

Lesson 1

Content: Subtraction of whole numbers without borrowing (regrouping)

Example 1:

Subtract:
```
  5 4 6
- 1 2 5
```

Solution
```
  5 4 6 - subtract ones 6-5= 1
- 1 2 5 - subtract tens 4-2=2
  4 2 1 - subtract hundred 5-1=4
```

Example 2:

Take away 2460 from 14693

Solution
```
TTh Th H T O
 1   4 6 9 3 - subtract ones 3-0 = 3
-    2 4 6 0 - subtract tens 9-6=3
 1   2 2 3 3 - subtract hundreds 6-4=2
               subtract thousand 4-2=2
              -subtract ten thousand 1-0=1
```

Activity.

Subtract the following

1. 4632 - 110

2. 9463
 - 8142

3. 569
 - 75

4. Takeaway 63 from 99

5. Subtract: 8946
 - 736

More on or addition of whole numbers.

Mary is 48years old . Her brother John is 14y older than her.

a) How old is John?

  4 8 years
+ 1 4 years
  6 2 years

b) Find their total age .

  4 8 years
+ 6 2 years
 110 years

2) Paul is year old . Sam is 19 years older than Paul.

a) How old is Sam .
  1 9
+1 2
  3 1years

b) Find their total age .

  1 2 years
+3 1 years
  4 3years

Activity

1) Charles is 20 years . Robert is 10 older than Charles.

a) How old is Robert?

b) Find their total age.

2) Fred is 33years old . Shafiq is 5 years older than Fred . a) How old is Shafiq?

b) Find their total age.

Week 6
Lesson 2
Content: Subtraction of whole numbers with regrouping.

Example 1.

subtract : 246
- 192

Solution

HTO

$^{1}\cancel{2}^{1}\cancel{4}6$ - subtract ones 6-2=4

- 1 9 2 - subtract tens (4-9)

0 5 4 10+4 =14-9=5

- Subtract hundreds 1-1=0

Example 2.

Subtract 254 from 932

HTO

$^{8}\cancel{9}^{12}\cancel{3}2$ - subtract ones 2-4= impossible

- 2 5 4 10+2=12-4=8

6 7 8 - subtract tens 2-5 10+2=12-5=7

-subtract hundreds 8-2=6

Activity.

Subtract the following.

1. 1 4 2
   - 1 3 4

2. 9 6 3
   - 8 9 9

3. 3 6 1
   - 1 8 3

4. 1 4 7
   - 6 8

5. 1 2 3 2
   - 4 8 9

6. 4 4 6 5
   - 3 8 7 9

Week 6
Lesson 3
Content: Evaluating facts on subtraction of whole numbers

Example 1:

What is the difference between 469 and 234?

Solution

  469
- 234

  235

Example 2:

Katamba s salary was sh.124500 and he was deducted sh. 1200. How much does he get now?

Solution

sh. 124500

sh. - 1200

sh. 123,300

Activity.

1. Take away 99 from 100.

2. Find the difference between 498 and 199.

3. Subtract 786 from 4356.

4. A man earns sh.45200 and his wife earns sh.12394. What is the difference between their salary?

5. In a certain school , there 1345 pupils and 7043 are boys. How many girls are there in the school?

6. By how much is sh. 1500 is similar than sh. 4500?

Week 6

Lesson 4

Content: Multiplication of numbers by one digit.

Example 1:

Multiply:
$$\begin{array}{r} 314 \\ \times\ \ 3 \\ \hline \\ \hline \end{array}$$

Solution. Method 1.

314 (Using repeated addition)

$$\begin{array}{r} 314 \\ \times\ \ 3 \\ \hline \\ \hline \end{array} \longrightarrow \begin{array}{r} 314 \\ 314 \\ +\ 314 \\ \hline 942 \\ \hline \end{array}$$

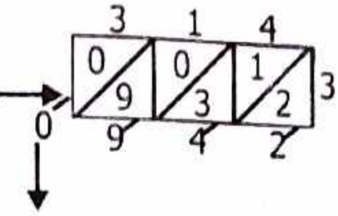
Method III using direct multiplication

```
 3 1 4
x    3
------
 9 4 2
```

Multiply 314 by ones.
314 x 3 = 942

Method IV using letice method

```
 3 1 4
x    3
------
```



= 942

Example 2.

Work out:
```
  1 8 4
x     4
-------

-------
```

Solution

```
 184
x  4
----
 736
```

-Using direct multiplication
-Multiply 184 by ones.
84 x 4 = 736

Method II

184 → 100 + 80 + 4
x 4 → x 4

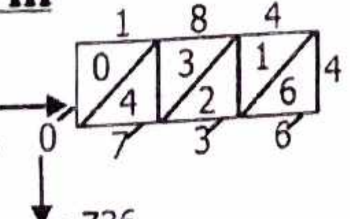
= 400+320+16

= 400
= 320
= +16

736

Method III

```
 1 8 4
x    4
------
```



= 736

Activity

Multiply the following numbers.

```
1.  4 1 3          2.  7 6 3
   x    3             x    6
   ------             ------

   ------             ------

3.  5 6 8          4.  7 3 3
   x    6             x    8
   ------             ------

   ------             ------

5.  3 2 3          6.  3 0 3
   x    9             x    9
   ------             ------

   ------             ------

7.  7 5 9          8.  9 9 5
   x    8             x    4
   ------             ------

   ------             ------
```

~~**Week 6**~~

Lesson 5

Content: Multiplication of numbers by 2 -digits.

Example 1.

Work out:
```
   6 5
x  1 0
------

------
```

Solution

```
 T 0
 6 5
x1 0
----
+0 0
6 5 0
-----
6 5 0
```

-Multiplying by ones 65x0=00.
-Multiply 65 by tens 65x10=650

Method 2.

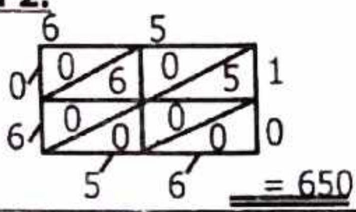
65 → 60 + 5
x1 0 → x 10

= 600 + 50

= 600
+ 50
650

Method 2.

65
x10
650

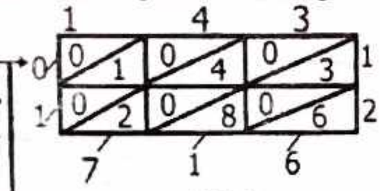


Example 2.

Work out: 1 4 3
x 1 2

Solution. **(Method 1.)**

143
x12



= 1716

(Method 2.)

143 = 100 + 40 + 3
x12 x 12

= 1200+480+36

= 1200
= 480
= + 36
= 1716

(Method 3.)

143 - multiply 143 by ones
x12 143x2 =286
286 143x10 = 1430
+1430
1716

Activity.

Multiply the following

1. 26 x 12

2. 431 x 12

3. 156 x 11

4. 145 x 13

5. 26 x 10

6. 99 x 13

7. 16 x 17

8. 135 x 11

Week 7
Lesson 1
Content: Evaluating facts or multiplication of whole numbers.

Example 1:
Find the product of 146 and 4.
Solution

```
 146
x  4
 584
```

Examples 2.
By selling a bunch of matooke, a seller got a profit of sh.3900. If he sold 12 bunches. How much did he get?
Solution

```
        sh. 3900
            x 12
            7800
          +3900
Profit=sh.46800
```

Activity.

1. Shafik is 10 years old. Mukasa is thrice the age of Shafik. How old is Mukasa now?

2. Juma paid sh.7800 for his lunch. How much will he pay for a week on lunch at the same cost?

3. Each of the schools in Wakiso district has 710 pupils. How many pupils are in the 5 schools?

4. A bus carries 28 passengers per trip. How many people will it carry in 16 trips?

5. What is the product of sh. 42000 and 12?

Week 7

Lesson 1

Content: Multiplication as repeated addition

Examples 1.

Work out 4x2 using repeated addition.

Solution

4x2 = 4 + 4
= 8

Examples 1.

Work out 4x3 using repeated addition.

Solution

4x3 = 4 + 4 + 4
= 12

Activity

Use repeated addition to workout the following.

Solution

1. 3x2

2. 4 x 6 =

3. 3 x 4 =

4. 2 x 8 =

5. 3 x 5 =

6. 9 + 9 + 9 = ______x ______

7. 10 + 10 +10 + 10 = ___x ___

8. 2 + 2 + 2 + 2 + 2 + 2 = __x__

Week 7

Lesson 3

Content: Division of whole number

Sub-Topic: Division

Example.

Workout : 12÷3

Solution

12÷3 =
12-3= 9 (1st subtraction)
9-3 = 6 (2nd subtraction
6-3 = 3 (3rd subtraction)
3-3 = 0 (4th subtraction)

∴ 12÷3 = 4

Example 2.

Divide 18÷6

18-6= 12 (1st subtraction)

12-6 = 6 (2nd subtraction

6-6 = 0 (3rd subtraction)

∴ 18÷6 = 3

Activity.

Workout the following using repeated subtraction.

1. 12÷2

2. 18÷3

3. 25÷3

4. 45÷9

5. 36÷9

6. 10÷2

7. 20÷4

8. 24÷8

Week 7
Lesson 4
CONTENT : Division of whole number without regrouping / borrowing using long division

Example 1.
Divide 370÷5
Soln.
Divide 370÷5

```
            074
        5 / 370  ───→ (3÷5)
(5x0) =     0          impossible
            37 ───→ (37÷5)
(5x7) =     35
             20 ───→ (20÷5)
(5x4) =      20
```

$\therefore$ 370 ÷ 5 = 74

Example 2.
Workout: 4804÷4=
Solution

```
            1201
        4 / 4804 ───→ (4÷4)
(1x4) =     4
            08 ───→ (8÷4)
(2x4) =      8
             00 ───→ (0÷4)
(0x4) =       0
              4 ───→ (4÷4)
(1x4) =       4
              0
```

$\therefore$ 4804 ÷ 4 = 1201

Activity.
Divide the following numbers using long division.
1. 124÷4

2. 375÷5

3. 921÷3

4. 1692÷2

5. 256÷4

6. 174÷3

Week 7

Lesson 5

CONTENT : More about division using long division.

Example 1.

Divide 24072÷3

Solution.

```
                08024
             4 ⟌24072 ──→ (2÷3) impossible
(0x3)  =        0↓
               24     ──→ (24÷3)
(8x3)  =       24↓
                00    ──→ (0÷3) impossible
(0x3)  =         0
                07    ──→ (7÷3)
(2x3)  =          6
                 12   ──→ (12÷3)
(4x3)            12
```

∴ 24072 ÷ 3 = 8024

Example 2:

Workout : 21350÷7

Solution

```
                03050
             7 ⟌21350 ──→ (2÷7) impossible
(0x7)  =        0
               21     ──→ (21÷7)
(3x7)  =       21↓
                 3    ──→ (3÷7) impossible
(0x7)  =         0↓
                35    ──→ (3÷7)
(5x7)  =        35
                  0   ──→ (0÷7) impossible
                  0
```

∴ 21350 ÷ 7 = 3050

Activity.

Workout the following.

1. 720 ÷ 8

2. 21350 ÷ 7

3. 34627÷9

4. 240727÷6

5. 728÷8

6. 7345÷5

Week 7

Lesson 6

CONTENT : Division of whole numbers with reminder

Examples 1.

Workout: 384÷5

denominator → 5

076 → Whole number

5 ⟌ 384 → (3÷5) impossible

(0x5) = 0

38 → (38÷5)

(7x5) = 35

34 → (34÷5)

(6x5) = 30

4 → (Reminder Numerator)

$\therefore$ 384 ÷5 = $76\frac{4}{5}$

Examples 1.

Divide: 38÷9

denominator → 9

04 → Whole number

9 ⟌ 38 → (3÷9) impossible

(0x9) = 0

38 → (38÷9)

(4x9) = 38

2 → (Reminder Numerator)

$\therefore$ 38 ÷9= $4\frac{2}{9}$

Activity.

Divide the following numbers using long division

1. $463 \div 2$

2. $46 \div 5$

3. $293 \div 3$

5. $423 \div 7$

Week 7

Lesson 7

CONTENT : Division by 2-digits (10)

Example 1.

Divided $650 \div 10$

Solution:

$650 \div 10$

$= \frac{65\cancel{0}}{1\cancel{0}}$ reduce the zeros

$= \frac{65}{1}$ $\quad = 650 \div 10 = 65$

Example 2.

Workout: $430 \div 10$

Solution:

$430 \div 10$

$= \frac{43\cancel{0}}{1\cancel{0}}$ reduce the zeros

$= \frac{43}{1}$ $\quad = 430 \div 10 = 43$

Activity.

Workout the following

1. 1200÷10
2. 460÷10
3. 570÷10
4. 900÷10
5. 340÷10
6. 550÷10
7. 6410÷10
8. 4310÷10
9. 200÷10
10. 100÷10

Week 8

Lesson 1

CONTENT : Evaluating on division of whole numbers.

Example 1.

Share 362 mangoes among 2 boys.

Soln.

$$
\begin{array}{rr|l}
 & 181 & \\
\cline{2-2}
 & 2\,\big)\,362 & \longrightarrow (3\div 2) \\
(1\times 2) = & 2 & \\
\cline{2-2}
 & 16 & \longrightarrow (16\div 2) \\
(8\times 2) = & 16 & \\
\cline{2-2}
 & 2 & \longrightarrow (2\div 2) \\
(1\times 2) = & 2 & \\
\cline{2-2}
 & 0 &
\end{array}
$$

$\therefore$ Each boy got 181 mangoes

Examples 2.
Divide 246 books among 3 schools.

Soln.

$$\begin{array}{rl} & \underline{082} \\ 3 \,\big)\, & 246 \longrightarrow (2\div3) \text{ impossible} \\ (0\times3) = & \underline{0\downarrow} \\ & 24 \longrightarrow (24\div3) \\ (8\times3) = & \underline{24} \\ & 6 \longrightarrow (6\div3) \\ (2\times3) = & \underline{6} \\ & 0 \end{array}$$

$\therefore$ Each school will get 82 books.

Example 3.
Joannah distributed 420 desks in the school in her country. How many desks did each school receive if there are 10 schools?

Solution.

$\frac{42\cancel{0}}{1\cancel{0}}$ reduce by 10.

= 42

Each school received 42 desks

ACTIVITY

1. A school bursar collected sh.42000 from 7 pupils. How much did each pupil pay?

2. Share 462 coins among 3 girls.

3. A man used 960 litres to drive a distance 10km. How many litres does he need to cover a distance of 1km?

4. Mr. Mwaje has 3 workers. He pays them a total of sh.36459 a day. How much did each worker receive?

5. Distribute 240 textbooks among 8 schools.

Week 8

Lesson 2

CONTENT : Finding average.

Sub-Topic: Average.

Finding average / mean

Example 1.

Find the average of 0, 4, 6, 5, 10

Solution.

$$\text{Average} = \frac{\text{sum of given data}}{\text{Number of given data}}$$

$$= \frac{0+4+6+5+10}{5}$$

$$= \frac{25}{5}$$

$$= 5$$

Example 2.

Workout the mean of 40, 50, 5 5

Solution

$$\text{Mean} = \frac{\text{sum of given data}}{\text{No of given data}}$$

$$= \frac{40 + 50 + 5 + 5}{4}$$

$$= \frac{100}{4}$$

$$= 25$$

Activity

1) Find the average of 0,2 and 4

2) Find the mean of 8,0,2, and 6

3) Find the average of 4,6,8 and 10 .

4. Find the mean of 20, 15, 35, and 30

5. Find the mean of 6,8,10,12 and 14.

Week 8

Lesson 3

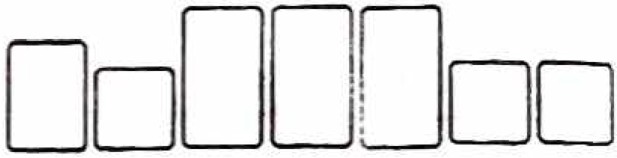
PATTERNS AND SEQUENCES

A pattern is an arrangement of geometrical shapes or number in an orderly manner.

- Sequences are patterns of shapes or numbers following a particular order.
- A sequence is obtained by applying some consecutive rule.
- Recognizing and naming common shapes.

Examples.

Recognise and name them.

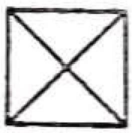
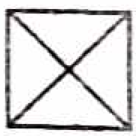


This pattern has 2 small squares, then 3 rectangles and then 2 small squares again.

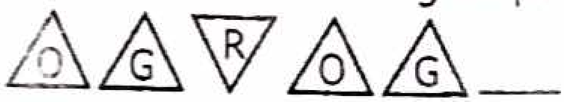
The missing shapes are 3 ectangles.

Activity.

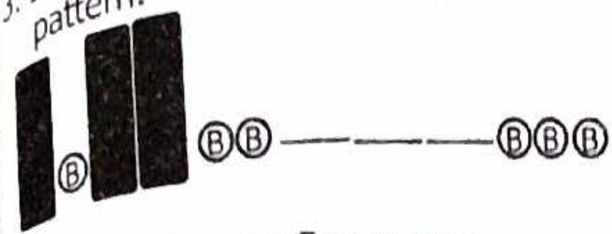
1. How many triangles are in the diagram?

a  b 

2. Draw the next missing shape.



3. Identify the missing shapes in the pattern.



4. Draw and name 5 common different shapes.

5. Make patterns and sequences using shapes.

Patterns and sequences formed by subtraction.

Examples.

1. You are going to keep on subtracting 3 to from a pattern.

a) 40-3 = 37
b) 37-3 = 34
c) 34-3 = 31
d) 31-3 =28
e) 28-3= 25
f) 25-3 = 22
g) [illegible] = 19
h) 16-3= 13
i) 13-3 = 10

Activity

Find the missing numbers.

1. 16, 15, 14, 13,____, ____, ____
2. 35, 30, 25,____, ____
3. 36, 31,____, ____
4. 28, 20, ____, ____
5. 25, 22, ____, ____
6. 50, 40, ____, ____, 10
7. 30, 28, ____, ____
8. ____, 14, 11, 8, 5, 2
9. ____, 21, 17, 13, 9, 5, 1

Patterns and sequences formed by addition.

Note:

We can use numbers instead of shapes to form patterns.

- These numbers follow each other in a pattern and we call this a sequence.

Examples.

1) Think of a number and keep adding 9. You will form a pattern.

2+9 = 11 2,11,20,29,38,47,____

11+9 = 20 What is the missing number?

20+9 = 29 Add 9+47 = 56

29+9= 38

38+9 = 47

The missing number 1556

2. **Find the missing numbers and fill in the boxes.**

3+3 = 6 3, 6, 9, 12, 15,____

6+3 = 9 15+3 =18

9+3 = 12 The missing number

12+3 = 15 is 18

Activity.

Fill in the missing numbers.

1. 2, 4, 6, 8, ____, ____, ____

2. 1, 3, 5, 7____, ____, ____

3. 8, 16, 24,____, ____

4. 6, 10, ____, ____

5. 10, 20, 30, ____, ____

CONTENT : Types of numbers

Topic: Number Pattern & sequences.

Even and odd numbers

Even numbers are numbers when divided by 2 leaves a reminder of zero. (0)

Examples; 0, 2, 4, 6, 8,10, 12, 14, 16, 18, 20, ...

Odd numbers are numbers when divided by 2 leaves a reminder of 1.(one)

Examples, 1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21, 23, 25, ...

Activity.

1. Divide the following numbers by 2 and tell whether its an even or odd number.

a) 8

b) 11

g) 12

c) 13

h) 15

d) 21

e) 48

f) 10

Week 8

Lesson 4

CONTENT : More about Even and odd numbers.

Example 1:

How many even numbers are there between 10 and 20?

Soln.

Even numbers between 10 and 20 = 12, 14, 16, 18.

. . Even numbers between 10 and and 20 are **4**.

Example 2:

How many odd numbers are there between 0 -10.

Odd numbers between 0 and 10 = 1, 3, 5, 7, 9.

.. **There are 5 numbers between 0 and 10.**

Activity.

1. List down all the fast 10 odd numbers.

2. How many even numbers are there between 15 and 30?

3. Find the first 4 even numbers in their order.

4. How many odd numbers are there between 5 and 35?

5. List down all even numbers between 100 and 150.

Week 8
Lesson 5
CONTENT : More about Even and odd numbers.

Finding the sum, difference and product of even numbers.

Example 1:
Find the sum of the first 6 even numbers.

Soln.

0, 2, 4, 6, 8, 10

Sum = 0 + 2+ 4 + 6 + 8 + 10
= 30

Example 2:
What is difference between the second and the 10th even numbers.

Solution

0 ,2 ,4 ,6,8, 10, 12, 14,16,18, 20,22

2nd 10th

Difference = 18 - 2
= 16

Examples 3.
Find the product of the 6th and 12th even number.

Soln.

0, 2, 4, 6, 8, 10, 12, 14, 16, 18, 20, 22, 24

6th 12th

product = 22
$\times 10$
00
+ 22
220

Activity.

1. Find the sum between the 1st and 4th even number.

2. Work out the difference between the 15th and 17th even number.

3. Find the product of the 2nd and 10th even number.

4. What is the sum between the 7th and 18th even number?

5. What is the difference between the 9th and 1st even number?

Week 8
Lesson 6
CONTENT : Find the sum difference and product of odd number.

Example 1.
What is the difference between the 2nd and 4th odd numbers?

Solution
1, 3, 5, 7, 9, 1
2nd 4th
Difference = 7 - 3 = 4

Example 2.
Solution.
Workout the sum between the 1st and the 10th odd number.
1, 3, 5, 7, 9, 11, 13, 15, 17, 19
1st 10th
sum = 19
+ 1
20

Example 3.

Find the product between the 9th and 5th odd number.

Solution.

1, 3, 5, 7, 9, 11, 13, 15, 17, 19, 21

Product = 17 × 9

$$\begin{array}{r} 17 \\ \times\ 9 \\ \hline 153 \\ \hline\hline \end{array}$$

Activity.

1. Workout the sum of the 3rd and 15th odd numbers.

2. Find the product between the 7th and 8th odd numbers.

3. What is the difference between the 3rd and the 18th odd number?

4. Find the sum between the 10th and 12th odd numbers.

5. Divide the 8th odd number by 3.

Week 8

Lesson 7

CONTENT : Counting and whole numbers.

- Counting numbers are numbers which start from one 1 up to infinite.
- Counting numbers are also known as Natural numbers.

Examples are 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 12................

Whole numbers : are numbers which starts from zero(0) upto infinite.

Note: Examples are 0, 1, 2, 3, 4, 5 6, 7, 8, 9, 10, ...

Note: Counting numbers starts from 1 upto infinite but whole numbers starts from 0 upto infinite

Activity.

1. Workout the list of the first 6 counting numbers.

2. Find by listing the first 10 whole numbers.

3. Fill in the missing whole numbers

a) ___, ___, ___, 3, 4, 5, ___, ___

b) 12, 13, ___, ___, 16, ___.

Week 9

Lesson 1

CONTENT : Number sequences

Note: There are two types of sequences; namely

1. Increasing sequence

2. Decreasing sequence

Increasing sequence:
This is a type of sequence which requires addition or multiplication to get the next number.

Examples.
Find the next number in the sequence below.
1, 3, 5, 7, ___, ___

Soln.

1, 3, 5, 7, 9, 11

+2 +2 +2 +2 +2 +2

Next no. $7 + 2 = 9$

Next no. $9 + 2 = 11$

Example 2.
Find the next two numbers in the sequence below.
1, 2, 4, 5, 7, 8, ___, ___

Soln.

1, 2, 4, 5, 7, 8, 10, 11

+2 +2 +1 +2 +1 +2 +1

Next no. $8 + 2 = 10$

Next no. $10 + 1 = 11$

Example 3.

Find the next number in the sequence below.

2, 4, 8, ____, ____

Soln.

2, 4, 8, 16, 32

$\times2$ $\times2$ $\times2$ $\times2$

Next no. $8 \times 2 = 16$

Next no. $16 \times 2 = 32$

Activity.

Find the next numbers in the sequences below

1. 1, 2, 3, ____, ____

2. 32, 34, 36, ____, ____

3. 3, 6, 12, 24, ____, ____

4. 1, 7, 49, ____, ____

5. 5, 7, 9, ____, ____

6. 15, 17, 19, 21, ____, ____

7. 36, 38, 40, 42, ____, ____

8. 100, 200, 300, ____, ____

Week 9

Lesson 2

CONTENT : Decreasing sequence.

Definition:

Decreasing sequence is a type of sequence which requires subtraction and division to get the next number in the sequence.

Examples 1.

Find the next number in the sequence below.

8, 6, 4, ____, ____

soln.

8, 6, 4, 2, 0

-2 -2 -2 -2

Next no. 4 - 2 = 2

Next no. 2 - 2 = 0

Example 2.
Find the next number in the sequence below.
21, 18, 15, ____, ____

Soln.
21, 18, 15, 12, 9

-3 -3 -3 -3

Next no. 15 -3 = 12

Next no. 12 - 3 = 9

Example 3.
Find the next number in the sequence below.
81, 27, 9, ____, ____

Soln.
81, 27, 9, 3, 1

÷3 ÷3 ÷3 ÷3

Next no. 9÷3 = 3

Next no. 3 ÷ 3 =1

Activity.

Find the next number in the sequences below.

1. 18, 15, 12, 9, ____, ____
2. 100, 50, ____, ____
3. 30, 28, 26, ____, ____
4. 44, 22, 11, ____
5. 32, 16, ____, ____
6. 140, 120, 100, ____, ____

Week 9

Lesson 3

CONTENT : Multiples of numbers.

Definition:

A multiple is a product of a given number

e.g 6 x 3 = 18 . **18 is a multiple of 6.**

Example 1.

Complete the multiple of 2 below

Soln.

1x2 = 2
2x2 = 4
3x2 = 6
4x2 = 8
5x2 = 10
6x2 = 12
7x2 = 14
8x2 = 16
9x2 = 18
10x2 = 20
11x2 = 22
12x2 = 24

1x5 =_____	1x6 =_____
2x5 =_____	2x6 =_____
3x5 =_____	3x6 =_____
4x5 =_____	4x6 =_____
5x5 =_____	5x6 =_____
6x5 =_____	6x6 =_____
7x5 =_____	7x6 =_____
8x5 =_____	8x6 =_____
9x5 =_____	9x6 =_____
10x5 =_____	10x6 =_____
11x5 =_____	11x6 =_____
12x5 =_____	12x6 =_____

ACTIVITY

Complete the multiple table below.

1x3 =_____	1x4 =_____
2x3 =_____	2x4 =_____
3x3 =_____	3x4 =_____
4x3 =_____	4x4 =_____
5x3 =_____	5x4 =_____
6x3 =_____	6x4 =_____
7x3 =_____	7x4 =_____
8x3 =_____	8x4 =_____
9x3 =_____	9x4 =_____
10x3 =_____	10x4 =_____
11x3 =_____	11x4 =_____
12x3 =_____	12x4 =_____

Week 9

Lesson 4

CONTENT : Listing multiples of numbers

Example 1.

List the first 20 multiples of 6.

solution.

1x6 =6	11x6 =66
2x6 =12	12x6 =72
3x6 =18	13x6 =78
4x6 =24	14x6 =84
5x6 =30	15x6 =90
6x6 =36	16x6 =96
7x6 =42	17x6 =102
8x6 =48	18x6 =108
9x6 =54	19x6 =114
10x6 =60	20x6 =120

The first 20 multiples of 6 are 6, 12, 18, 24, 30, 36, 42, 48, 54, 60, 66, 72, 84, 90, 96, 102, 108, 114, 120

Activity.

List the first 10 multiples of the following numbers.

1. 6 are ....
2. 7 are ....
3. 8 are ....
4. 9 are ....
5. 10 are ....
6. 11 are ....
7. 12 are ....
8. 60 are ....

Week 9

Lesson 5

CONTENT : More about multiples of numbers.

Example 1.

List the multiples of 2 between 10 and 20.

solution.

1x2 = 2	6x2 = 12
2x2 = 4	7x2 = 14
3x2 = 6	8x2 = 16
4x2 = 8	9x2 = 18
5x2 = 10	10x2 = 20

The multiples of 2 between 10 and 20 are 12, 14, 16, 18.

Example 2:

Find the multiples of 12 less than 40.

Solution

1x12 =12

2x12 =24

3x12 =36

4x12 =48

The multiples of 12 between less than 40 are 12, 24, 36.

ACTIVITY

1. List the multiples of 2 less than 15.

2. List all the multiples of 18 less than 120.

3. List the multiples of 7 less than 50.

4. List all the multiples of 8 between 20 and 30

Week 9

Lesson 6

CONTENT : Finding common multiples of numbers.

Example 1.

Find the common multiples of 3 and 6.

solution.

M3= 3, 6, 9, 12, 15, 18, 21, 24, 30, 33, 36, 39, 42, 45

M6= 6, 12, 18, 24, 30, 36, 42, 48, 54, 60

Common multiples are 6, 12, 18, 24, 30.

Examples.

Find the common multiples of 7 and 9.

M7= 7, 14, 21, 28, 35, 42, 49, 56, 63, 70, 77, 84 91, 98, 105, 112, 119 126

M9 = 9, 18, 27, 36, 45, 54, 63, 72, 81, 90, 99, 108 117, 126

Common multiples are 63, 126....

Activity.

Find the common multiples of the following pairs of numbers.

1. 2 and 3

2. 8 and 9

3. 4 and 5

. 3 and 5

5. 5 and 7

6. 6 and 7

7. 10 and 11

8. 6 and 8

Week 9

Lesson 7

CONTENT : Finding lowest common multiples (LCM) of numbers.

Example 1.

Find the Lcm of 2 and 3

solution. (Using multiples of numbers)

M2 = 2, 4, 6 8, 10, 12.

M3= 3, 6, 9, 12, 15,

Common multiples = 6 , 12.......

.. LCM of 2 and 3 is 6.

Method II using a table.

2	2	3
3	1	3
	1	1

Their products is the Lcm of 2 and 3 $= 2 \times 3$

$= 6$

Example 2.

Find the LCM of 6 and 7

Solution

M6 = 6, 12, 18, 24, 30, 36, 42, 48, 54, ...

M7 = 7, 14, 21, 28, 35, 42, 49, 56, ...

Common multiples 42...

The LCM of 6 and 7 is 42

Method 2.

2	6	7
3	3	7
7	1	7
	1	1

$= 2 \times (3 \times 7)$

$= 2 \times 21$

$= 42$

Activity.

Find the LCM of the following numbers;

1. 5 and 4

2. 2 and 6

3. 3 and 2

4. 6 and 4

5. 8 and 12

6. 10 and 4

7. 6 and 8

8. 9 and 6

Week 10

Lesson 1

CONTENT : Finding factors of numbers

Note. A factor is a number that divides the given number completely without any reminder.

Example 1.

Find the factors of 6.

Solution. (Method 1)

$6 \div 1 = 6$

$6 \div 2 = 3$

$6 \div 3 = 2$

$6 \div 6 = 1$

$F6 = \{1, 2, 3, 6\}$

Example 2:

Find the factors of 12.

Solution. Method 2.

$12 \times 1 = 12$

$6 \times 2 = 12$

$4 \times 3 = 12$

$F12 = \{1, 2, 3, 4, 6, 12\}$

Activity

Find the factors of the following numbers

1. 6
2. 16
3. 14
4. 24
5. 36
6. 44
7. 20

8. 10

Week 10

Lesson 2

CONTENT : More about factors

Example 1.

Complete the table below and the find the factors of 12.

Factors	Multiple
1x12	12
2x6	12
3x4	12

Factors of 12={1, 2, 3, 4, 6, 12)

Activity.

Complete the tables below by finding the factors of the following numbers.

1.

Factors	Multiple
1x_____	10
2x _____	10

Factors of 10 are ={__, __, __}

2.

Factors	Multiples
____x____	48
____x____	48
____x____	48
____x____	48
____x____	48
____x____	48
____x____	48

Factors of 48 are ={____, ____, ____, ____, ____, ____, ____, ____, ____, ____}

Factors	Multiples
____x____	30
____x____	30
____x____	30
____x____	30

Factors of 30 are {____, ____, ____, ____, ____, ____, ____, ____, ____}

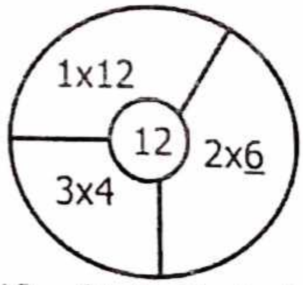
Week 10

Lesson 3

CONTENT : More about factors and multiples

Example 1.

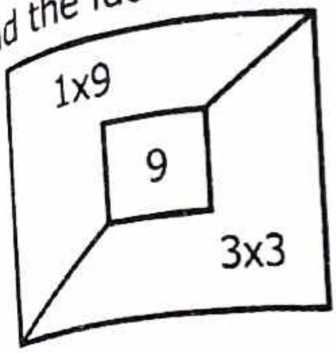
Find in and list the factors of 12.



F12 ={1, 2, 3, 4, 6, 12}

Example 2.

Find the factors of 9.

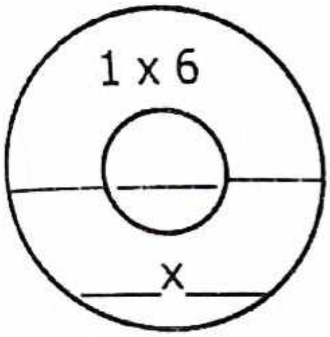


F9 ={ 1, 3, 9}

Activity.

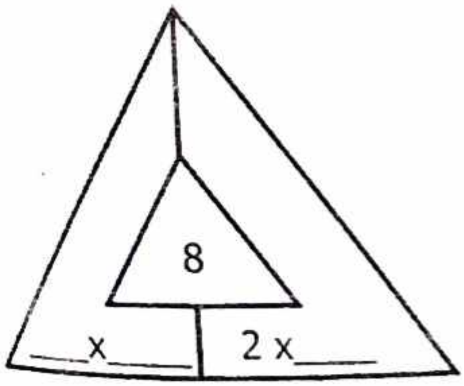
Fill in and complete to find the factors of the given multiples.

1.



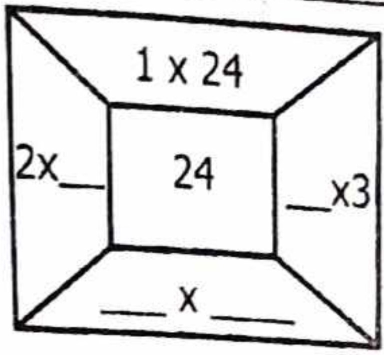
F6 ={ ______, ______, ______}

2.



F8 ={ ____, ____, ____, ____}

3.



F24 ={___, ___, ___, ___, ___}

Week 10

Lesson 4

CONTENT : Magic squares

Example 1.

Complete the table below to answer the questions that follow.

7	a	5
b	4	c
3	d	1

1. Find the magic sum.

Solution:

Sum = 7 + 4 + 1

= 12

2. Workout the value of the unknowns.

Solution.

for a

a + 7 + 5 = 12

a + 12 = 12

a + 12 - 12 = 12 - 12

a = 0

for b

$b + 7 + 3 = 12$
$b + 10 = 12$
$b + 10 - 10 = 12 - 10$
$b = 2.$

for c

$c + 1 + 5 = 12$
$c + 6 = 12$
$c + 6 - 6 = 12 - 6$
$b = 6.$

for d

$d + 1 + 3 = 12$
$d + 4 = 12$
$d + 4 - 4 = 12 - 4$
$d = 8$

Activity.

In the following tables below , find the sum and the value of the unknowns.

1.

2	a	6
b	5	c
4	d	8

2.

x	2	6
4	6	y
z	11	3

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TOPIC 1: SET CONCEPT

A set is awell- defined collection of elements or members.

Symbols used in sets.

$=$ $\longrightarrow$ Equal sets.

$\neq$ $\longrightarrow$ un equal sets.

$\cup$ $\longrightarrow$ Union of sets.

$\cap$ $\longrightarrow$ Intersection sets.

$\emptyset$ $\longrightarrow$ Empty set.

$\leftrightarrow$ $\longrightarrow$ Equivalent sets.

$\not\leftrightarrow$ $\longrightarrow$ Non-equivalent sets.

ε $\longrightarrow$ Universal sets.

$\subset$ $\longrightarrow$ Proper subset

$\subseteq$ $\longrightarrow$ Is a subset of

$\not\subset$ $\longrightarrow$ Is not a subset of

$\supset$ $\longrightarrow$ Proper superset

$\supseteq$ $\longrightarrow$ Is a superset of

$\not\supseteq$ $\longrightarrow$ Is not a superset of

$\in$ $\longrightarrow$ Is a member of

$\notin$ $\longrightarrow$ Is not a member of

C' $\longrightarrow$ complement of sets

Types of sets.

1. Empty sets / Null / Void.
2. Equal sets and unequal sets.
3. Joint sets / disjoint sets.
4. Equivalent sets and non-equivalent sets.
5. Intersection and union of sets.
6. Finite and infinite sets.
7. Even and odd sets.

1. Empty sets (Null sets)

An empty set is a set without any element.
Another name for empty set is Null set.
A symbol for empty set = { } or $\emptyset$

An empty set can also be called void set.

$\emptyset$

Examples of void sets.

1. K is a set of goats that can lay eggs.
2. A is a set of phones which walk.
3. C is a set of boys who are twice the age of their parents.

Describing members in empty sets.

Example 1.

How many members are in set K?

K

- There is no element in K. K is an empty set.

Example 2.

List and count elements of the sets below.

a) Set X = { all odd numbers less than 5)

b) Set P = { Vowel letters}

Solution:

Set X= { 1, 3}

Set P = { a, e, i, o, u }

n(X) = 2

n(P) = 5

Activity

State whether **empty** or **not empty**.

1. Set K = { days of the week}.

   ....

2. Set F = { animals which can fly}.

   ....

3. Set A ={ mothers who give birth to young ones}.

   ....

4. Set B={colours of our national flag}.

   ....

5. List and count the members of the sets below.

   a) Set M ={ months of the year}.

   ....

   ....

   b) Set L = { vowel letters}.

   ....

   ....

6. Write the set symbol for a null set.

....

7. What is an empty set?

....

....

....

8. Give any two examples of empty sets.

i) ....

ii ....

EQUAL SETS

Two sets are said to be equal if they have exactly the same number of elements of the same type.

Equal sets are sets that have the same type and the same number of members.

Symbol for equal sets is =.

Note: When two sets have elements not the same, we say unequal sets i.e

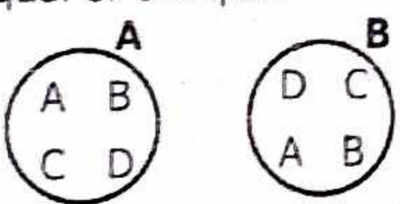
Set **A** = {1, 2, 3} and **B**={ 1, 2}.

We write set A is unequal set to B

or A $\neq$ B

Examples.

1. Describe the pairs of sets below as equal or unequal.



A and B have exactly the same members.

A is equal to B. **A=B**

2. Set A ={p, q, r, s , t} and Set N={r, s, t, p}

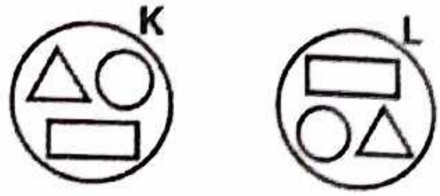
Set A is unequal to set N.

or A $\neq$ N

Activity

Write equal or not equal.

1.



....

2. T ={a, b, c, d} and S={a, d, c, b}.

....

3. P={5, 4, 6} and set N={a, b, c}.

....

4. G ={1, 2, 3} and B ={2, 1, 3}.

....

5. Write true or false in each of the following.

i) Equal sets have same number of elements.

....

ii) All equal sets have exactly the same elements.

....

iii) Sets with different numbers of elements are not equal.

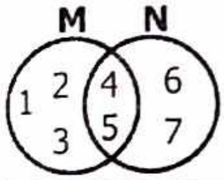
....

Joint And Disjoint Sets.

Joint sets have some common members.
Disjoint sets are sets which have no common members at all.

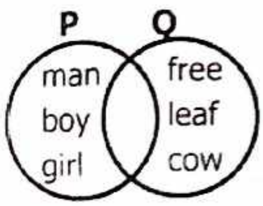
Examples.

1. Set M = {1, 2, 3, 4, 5}
   N = {4, 5, 6, 7}



$M \cap N = \{4,5\}$
Set **M** and **N** are Joint sets.

2. Set P = { man, boy, girl,}
   Q = {tree, leaf, cow}



PnQ = { }
P and Q are disjoint sets.

Activity

Write joint or disjoint.

1. M = {a, b, c} , N = {b, a, g}

   ....

2. K = { 1 , 2 , 3 } , T = {8, 4, 1}

   ....

3. B = {□, ○, △} = {▱, □}

   ....

4. T = {m, o, n, e, y}
   K = { c, l, o, t, h, e, s}

   ....

5. P = { O, D, E} K = {P, O, T}

   ....

6. E = {1, 3, 5, 7 } , V = {0, 2, 4, 6}

   ....

Equivalent and non -equivalent sets.

Equivalent sets are sets with the same number of elements but with different elements.
The set symbol for equivalent sets is $\longleftrightarrow$ or $\equiv$

Non equivalent sets have different types and numbers of elements or members.
The set symbol for non-equivalent sets is $\not\longleftrightarrow$ or $\not\equiv$

Examples.

Given that;

1. K = {b, c, d} and L = {c, d, b}
   Set K is equivalent to set L;
   $K \longleftrightarrow L$

2. If set A = { 1, 2, 3, 4} and
   B = { x, y, w, z }
   Set A is equivalent to set B;
   $A \longleftrightarrow B$

3. B = {x, y, w, z} and A = {1, 2, 3}
   B is non equivalent to set A.
   $B \not\longleftrightarrow A$

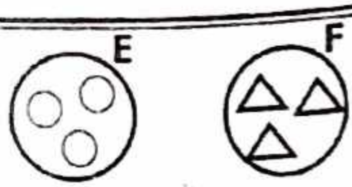
Activity

Given the following pairs of sets which ones are equivalent and which ones are not?

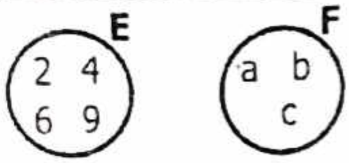
1.

A	B
4, 6, 8	3, 5, 7

2.



3.



4. T ={star, moon, ball}
N = { square, pencil }

5. M ={k, l, m, n }
N = { a, b, c, d }

6. W ={4, 6, 3, 8 }
Z = {first 4 ABC letters}

7. Q ={4, 6, 3, 8 }
V = {m, e, t, a, l}

Intersection Sets And Union of Sets.

Intersection of sets are sets formed by the common elements got from two or more given sets.

∩ is the set symbol for intersection.

When two or more sets are joined together they form a Union of set.

Examples.

1. If set A ={1, 2, 3} and
Q ={2, 4, 6}
Set AnQ = {2}

2. Given that A={p, q, r, s, t} and
B ={p, r, s, t}. Find n(AnB)
Set AnB = {p, r, t, s }
n(AnB) = 4 members.

3. If P ={1, 2, 3} and R = {5, 6, 8}
Find PUR.
PUR = {1, 2, 3, 5, 6, 8}

4. Given K = { a, b, c} and
M={b, a, g}
How many members are in set KUM?
Set KUM ={a, b, c, g}
n(KUM) = 4 members.

Activity

1. Given that A={a, e, i, o, u} and
B={e, i, o, u, f} .
Find i) AnB

ii) AUB

ii) X = {2, 3, 4, 6} and Y={0, 2, 7, 9}

iii) Z = {a, b, c, d} and S={f, i, v, e}

2. Write the intersection of these sets
a) Set L ={u, g, a, n, d, a}
M ={d, u, b, a, I}

b) G={triangle, circle, star }
H = { square, oval, book}

c) D = {water, milk, soup, juice}
E = {soup, water, porridge}

3. N={a, e, i, o, u} and K= { }.
Find set NUK.

4. Use the symbol U to form the union of sets of;
(i) M ={h, i, k, g, j} and
N={d, e, f, g, h}

Finite and Infinite sets.

Finite sets are sets with a limited number of elements.

Note: The elements have an end.

Examples.

1. Give set P ={Factors of 6}
P = {1, 2, 3, 6}.

b) B ={Multiples of 2 less than 10}
B={2, 4, 6, 8}

Infinite sets.

These are sets with a limitless number of items.

NB: Elements have no end or can not be listed all.

Examples.

1) A={even numbers}
A={0, 2, 4, 6, 8, 10,--------}

2. K ={All odd numbers}
K={1, 3, 5, 7, 9, 11,-------}

When listing elements of an infinite set, we write 3 dots (-----) at the end to show that the list is still continuing.

Activity

State whether infinite or finite sets.

1. A = {Multiples of 6 below 30}

2. B={Odd numbers between 6 and 15}

3. C = {all composite numbers}

4. D = {whole numbers above 10}

K = {Vowel letters}
P = {Factors of 36}

Even sets.

These are sets that have an even number of elements.

Odd sets.

Odd sets are sets with an odd number of members.

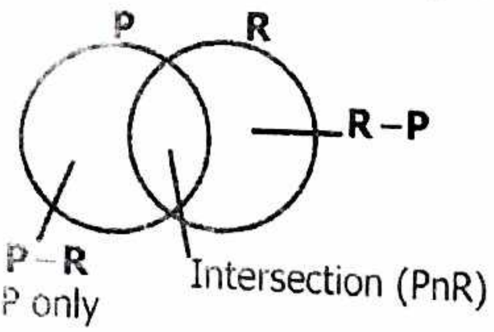
Remember.

Even numbers are numbers exactly divisible by 2 e.g. 0,2,4,6,8,..

Odd numbers are numbers that are not exactly divisible by 2 e.g. 1,3,5,7, and they give a reminder.

VENN DIAGRAMS

Naming parts on venn diagrams.

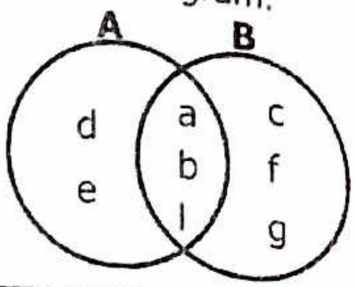


Drawing venn diagram to find intersection and union of sets.

Example.

If A ={a, b, l, d, e} and
B={a, b, l, c, f, g}

a) Represent the above information on a venn diagram.



b) List members of

i) AnB

A∩B = {a, b, l}

ii) A only

A only = { d, e}

iii) n(B) only

B only = {c, f, g}

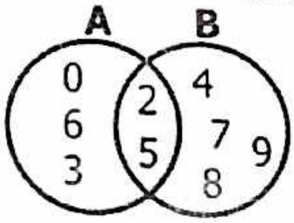
n(B) = 3

iv) AUB

AUB ={a, b, c, d, e, f, g, l}

Activity

1. List the elements of



i) A

ii) B

iii) AnB

iv) A - B

v) B - A

vi) n(BUA)

2. Given that M={a, b, c, d} and
N= {a, b, e, g, h},
Draw a venn diagram and show the elements.

3. Draw a venn diagram to represent the information below.

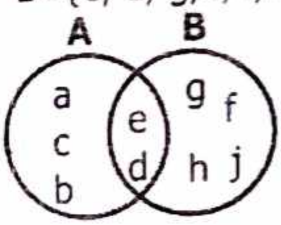
a) Set S ={1, 2, 3, 4, 5, 6, 7}
K = { 2, 4, 6, 8}

b) Set P ={0, 2, 4, 6, 8}
Q ={1, 3, 5, 7, 9}

Difference of sets.

Examples.

A={ a, b, c, l, d, e}
B={e, d, g, f, l, h, j}



i) A-B ={a, b, c} or B'

ii) B-A ={g,i, f, h} or A'

iii) n(B-A)= 4   iv) n(A-B) = 3

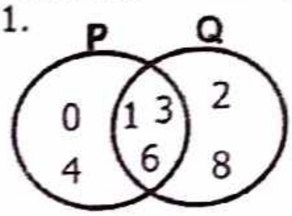
Note: A-B means members in set A. only but not in set B or (B complement) B'.

B-A means members in set B only but not in A (A complement) A'
B' = (a, b, c}
A' = {g, h, j}

Activity

Use the following sets to find.

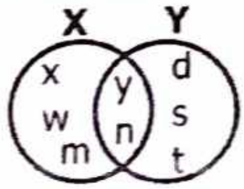
1.



(i) P' ______________

(ii) Q' ______________

2.

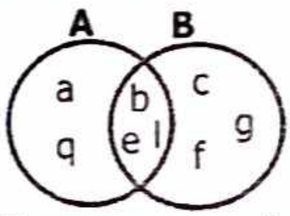


x - y ....

(xny)' ....

x' ....

3.



How many members are in

(i) B

(ii) B - A

(iii) A - B

SUB SETS

A subset is a small set found in a big set.

Set symbol for subset is $\subseteq$ and not subset is $\not\subseteq$.

Note: Universal set is a set that contains other smaller sets.
Universal set is a subset of itself though not a proper subset.

Examples.

1. P ={1, 2, 3, 4, 5, 6} ,
   K ={2, 4, 6} , E={1, 3, 5}
   Q ={9, 10}

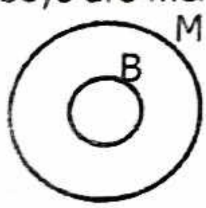
P =ε {P is a universal set of K and E}

E is a subset of P (E $\subseteq$ P).

K is a subset of p (K $\subseteq$ P)

Q is not a subset of P{Q $\not\subseteq$ P}

2. Draw a venn diagram to show that all boys are men.



Activity

1. Use diagrams to show the following

i) A $\subseteq$ B

ii) C $\subseteq$ D

iii) K $\subseteq$ L

2. Draw a venn diagram to show that all girls are female.

Listing subsets

1. Subsets can be identified by the use of the formular and listing.
   The formular is 2^n where n is the number of elements.

N.B: An empty set is a subset of every set.

Examples.

Set K ={a, b, c}

(i) Sub sets of K = { }, {a, b, c} , {a, b} , {b, c} , {a}, {b}, {c} , {a, c}

(ii) n(C) of K = 8 subsets.

Activity

List the following subsets.

1. Set P ={1,3}

2. Set M = {t, y, z}

3. Set y ={7, 6, 8}

Finding the number of subsets.

When finding the number of subsets apply the formula;
Number of subsets = 2^n
where n stands for the number of elements in the given set.

Examples.

Find the number of subsets in each set below.

a) M = {1, 2, 3}
Number of subsets $=2^n$
$=2^3$
$=(2\times2)\times2$
$= 4\times2$
= 8 subsets.

b) $n(A) = 5$

Number of subsets $= 2^n$

$= 2^5$

$= (2 \times 2)(2 \times 2) \times 2$

$= (4 \times 4) \times 2$

$= 16 \times 2$

$= 32$ subsets

c) $n(K) = 0$

Number of subsets $= 2^n$

$= 2^0$

$= 1$

Note: Any number to the power of zero is equal to one (1) apart from zero.

Activity

Calculate the number of subsets in each set below.

1. K = {a, e, i, o, u}

2. P = {1, 7, 4}

3. W = { 0 }

4. L = {8, 9}

5. T = { 9 }

6. $n(A) = 4$

7. $n(B) = 0$

8. $n(P \cup Q) = 7$

PROPER SUBSETS.

These are subsets of a given set excluding the union set itself.

The symbol "$\subset$" stands for "proper subset of".

Number of proper subsets $= 2^n - 1$ where n is the number of elements.

Examples.

1. P={1, 2, 3}

Number of proper subsets $= 2^n - 1$

$= 2^3 - 1$

$= 2 \times 2 \times 2 - 1$

$= 8 - 1$

$= 7$

2. V = { }

Number of proper subsets $= 2^n - 1$

$= 2^0 - 1$

$= 1 - 1$

$= 0$

Activity

Find the proper subsets of the following sets.

1. L = road, hospital, church

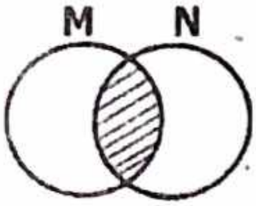
2. A = {Mali, Togo}

3. B= {Kenya, Uganda, DRC, USA}

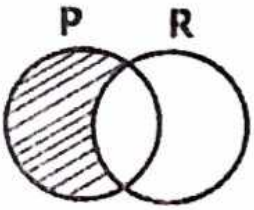
4. $n(P \cup R) = 5$

Shading and describing shaded regions.

a) Describe the shaded parts of the Venn diagrams below.



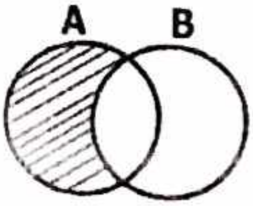
MnN



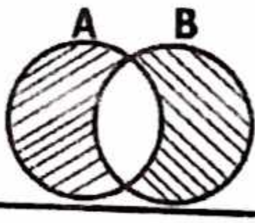
P - R / P only

2. Shade A - B

i)



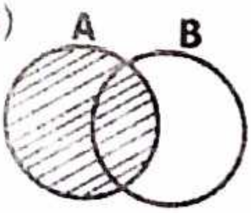
ii) (XnY)'



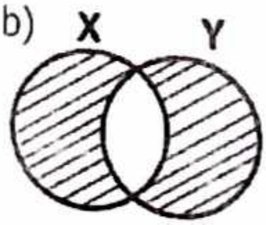
Activity

1. Describe the shaded regions of the given Venn diagrams.

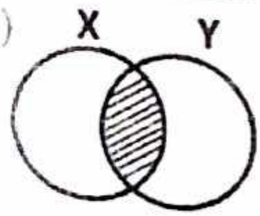
a)



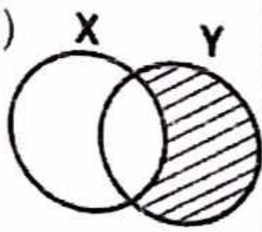
b)



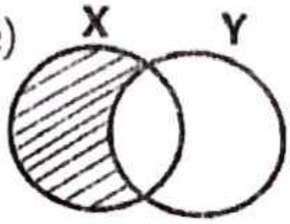
c)



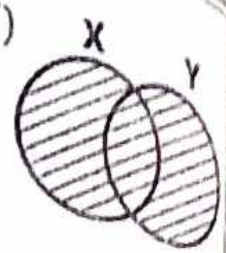
d)



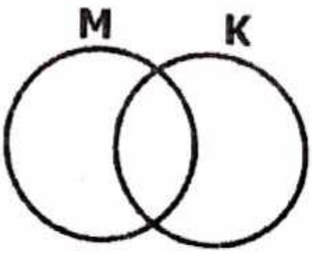
e)



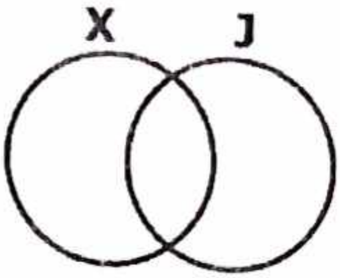
f)



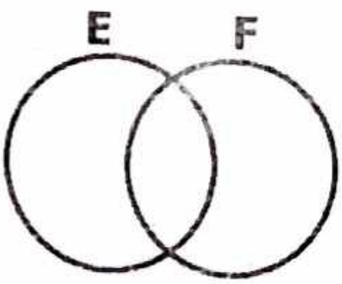
2. Shade set M only.



3. Shade X - J



4. Shade E U F



PROBABILITY IN SETS

Probability is the measure of chances.

Tossing a coin

When tossing a coin we get either head or tail.

Example

When you toss a coin, what is the probability of a head showing up?

Solution.

Sample space = {head, tail } = 2

Number of events = {head}

$$P = \frac{n(E)}{n(S)} = \frac{1}{2}$$

$$P = \frac{1}{2}$$

Probability of different items.

Examples:

There are 10 pencils in a tin, 3 of them are red and the rest are black. What is the probability of picking a black pencil randomly?

Solution:

Sample space = {10}

No. of events = Black = {10-3} = 7

$$P = \frac{n(E)}{n(S)} \qquad P = \frac{7}{10}$$

Activity

1. What is the probability of a tail appearing on top when you toss a coin?

2. What are chances of getting the following on a die.

a) 3

b) 4

iii) 5

iv) an odd number

v) an even number

3. In the basket, there are 4 ripe bananas and 6 raw bananas. Find the probability of getting a raw banana in the basket?

Probability using days of the week.

1. Annet will visit her friend next week. Find the probability that she will visit on a day starting with letter "s".

Solution.

$$\text{Probability} = \frac{DC}{TC}$$

$$= \frac{\text{Sat, Sun}}{\text{Mon, Tue, Wed, Thur, Fri, Sat, Sun}}$$

$$P = \frac{2}{7}$$

2. A teacher will eat fish next week, what is the chance that he will eat it on Wednesday?

Solution.

$$\text{Probability} = \frac{DC}{TC}$$

$$P = \frac{\text{Wed}}{\text{Mon, Tue, Wed, Thur, Fri, Sat, Sun}}$$

$$P = \frac{1}{7}$$

Activity

1. It is likely to rain next week. Find the probability that it will rain on a day starting with letter "T".

2. I want to visit USA next week. What is the probability that I will go there on Sunday?

3. What is the probability that we shall eat beef on Friday next week?

4. Martha will go to Kampala next week. What is the probability that she will go there on Monday and Tuesday?

Probability using a dice.

A dice is rolled and tossed at once. Find the probability that;

i) an even number shows up.

$$\text{probability} = \frac{EC}{TC}$$

$$P = \frac{2,\ 4,\ 6}{1,\ 2,\ 3,\ 4,\ 5,\ 6}$$

$$P = \frac{3}{6}$$

ii) prime number appears on top.

$$\text{probability} = \frac{EC}{TC}$$

$$P = \frac{2,\ 3,\ 5}{1,\ 2,\ 3,\ 4,\ 5,\ 6}$$

$$P = \frac{3}{6}$$

Activity

Kisale rolled and tossed a dice. Find the probability that;

1. A Multiple of 3 appeared on top.

2. A number less than 6 showed up.

3. Multiple of 2 appeared on top.

4. an odd number showed up.

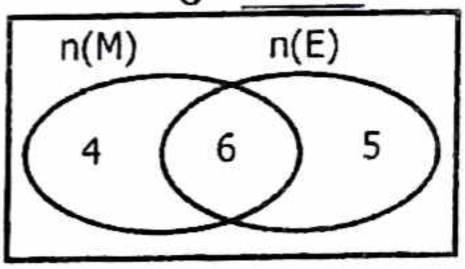
5. Composite number showed up.

Using Venn diagrams to find items.

Example.

1. The Venn diagram below shows the number of pupils in P.5. Use it to answer the questions that follow.

$\varepsilon =$ ______



b) Find the total number of pupils who liked Maths.

4 + 6 = 10 pupils.

c) How many pupils liked English?

6 + 5 = 11 pupils.

d) How many pupils are in Primary five?

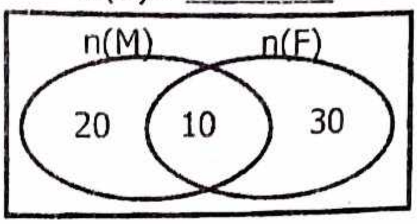
4 + 6 + 5

=15 pupils.

Activity

In a group of 60 people, 30 of the people liked eating fish and others meat as shown below.

n(ε)= ________



a) How many people liked both fish and meat?

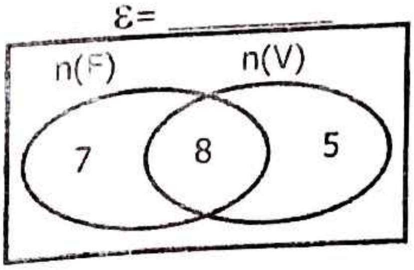
b) Find the total number of people;

i) who took meat

ii) who took fish

2. The venn diagram below shows the number of pupils who preferred different games, football and volley ball.

ε= ________



Find the number of pupils who preferred;

i) foot ball

ii) Volley ball

b) How many pupils are in that class?

TOPIC 2: WHOLE NUMBERS

ROMAN NUMERALS

General Roman numerals.

Roman Numeral	I	V	X	L	C	D	M/T
Hindu - Arabic	1	5	10	50	100	500	1000

Roman Numerals Obtained by Repeated addition.

Roman Numeral	II	III	VI	VII	VIII	XX	XXX	LX	LXX	LXXX	CC	CCC	DC
Hindu - Arabic	2	3	6	7	8	20	30	60	70	80	200	300	600

Roman Numerals Obtained by Subtraction.

Roman Numeral	IV	IX	XL	XC	CD	CM
Hindu - Arabic	4	9	40	90	400	900

Expressing Hindu - Arabic numerals as Roman Numerals.

Examples

Change 25 into Roman Numerals.

Solution

25 = 20 + 5
= XX V
= XXV

Change 45 into Roman Numerals.

Solution

145 = 100 + 40 + 5
= C XL V
= CXLV

Activity

Express the following as Roman numerals.

1. 31
2. 89
3. 150
4. 162
5. 200
6. 54
7. 39
8. 28
9. 52
10. 144

Changing Roman Numerals to Hindu Arabic numerals.

Examples

1. Express XXXIII as Hindu - Arabic.

Soln

XXXIII = XXX III
= 30 + 3
= 33

2. Change CXLV into Hindu - Arabic Numerals.

Soln

CXLV = C XL V
= 100 + 40 + 5
= 145

Activity.

Express the following Roman numerals as Hindu Arabic numerals.

1. 96	6. 143
2. 24	7. 155
3. 18	8. 290
4. 38	9. 74
5.46	10. 88

Evaluating facts on Roman Numerals.

Examples.

1. John was given XL litres of milk and his sister Jane was given XXV litres of milk. Find how many litres of milk they had altogether in Hindu - Arabic Numerals.

soln

John had XL = 4 0 litres
Jane had XXV = + 2 5 litres
6 5litres of milk

2. If Peter was XI years old last year 2023. Find his age in Hindu Arabic Numerals.

soln

XI = X I
= 10 + 1
= 11 years old

Activity.

1. It is now XII minutes past 11 o'clock in the morning. Express this in figures.

2. Rihana is X years old and Asha is XIV years old. Find their total age in Roman numerals.

3. Write what must be added to XXXIV litres of water to get LXIII litres of water in Hindu Arabic Numerals.

NOTE: Roman Numerals are only and only expressed in Capital letters.

PLACE VALUES OF WHOLE NUMBERS.

Place value is a position of a digit in a given number.

Each digit in a number has its own place value, the place value(S) determines the position of the digit.

Remember place values are given from right to left order starting with ones, tens, hundreds etc. And they must end with letter "S" or "ths" for decimals.

Examples;

1. Write the place value of each digit in 123456.

1 2 3 4 5 6

6 – Ones
5 – Tens
4 – Hundreds
3 – Thousands
2 – Ten thousands
1 – Hundred thousands.

2. State the place value of 7 in 37401.

37 401

7 – Thousands

The place value of 7 is thousands.

Activity

1. Write the place value of each digit in the following.

a) 203

b) 4076

c) 75664

d) 813 475

3. What is the place value of 7 in 673?

4. Which digit is in hundreds in 2643?

5. Give the place value of the underlined digit in the following.

a) **2**1438

b) **3**65402

c) 415300

Values of digit up to 999 999

The value of a digit depends on its place value. Multiply each digit by its place value to get its values.

Examples.

Find the value of each digit in the number 648, 205?

Number	Place value
6	Hundred thousands
4	Ten thousands
8	Thousands
2	Hundreds
0	Tens
5	Ones

Value

6 x 100,000 = 600,000
4 x 10,000 = 40,000
8 x 1000 = 8,000
2 x 100 = 200
0 x 10 = 0
5 x 1 = 5

2. Find the sum of the value of 5 and the value of 7 in 65978.

Solution:

Value of 5 = Digit x place value
5 x 1000 = 5000

Value of 7 in 65978
65978

value = Digit x place value
7 x 10
= 70

Sum = 5,000 + 70
= 5,070

Activity

1. Find the value of each digit in each of the following.

(i) 245

(ii) 999 99

2. Find the sum of the value of 2 and the value of 6 in 42360.

3. Calculate the value of 6 in the following.

a). 601

b). 796

More about values of digits.

Examples

1. Find the value of 4 tens.

Soln

Value = Digit x place value
= 4 x 10
= 40

2. What is the value of 146 hundreds?

Value = Digit x place value
= 146 x 100
= 14600

Activity

Find the value of the following numbers.

1. 3 ones

2. 4 ones

3. 7 hundreds

4. 15 thousands

5. 17 tens

6. 1154 ones

Reading and Writing numbers in words.

-To read and write a number in words follow these steps;
- Put the digits of the given number on a place table.
- Write the value of a number in each period in words.

Example.

1. Write 9634 in words.

Solution.

Thousands			Units		
H	T	O	H	T	O
		9	6	3	4

9634 = **Nine thousand, six hundred thirty four.**

2. 60781

Thousands / Units					
H	T	O	H	T	O
	6	0	7	8	1

60781 = **Sixty thousand, seven hundred eighty one.**

Activity

Write the following in words.

a) 546

b) 3246

c) 1,010

d) 202

e) 101

To write number words in figures.

- Draw a place value table showing periods.
- Put the number word each period, write a 3 digit number of each period.
- Add the values of digits in all the periods.

Examples.

Write four hundred fifty four thousand, one hundred sixty one in figures.

Thousands	**Units**
Four hundred fifty four fifty four thousand	one hundred sixty one
454	161

$$= \begin{array}{r} 454,000 \\ +\quad 161 \\ \hline 454,161 \end{array}$$

2. Write the single numeral for 24 thousands 7 hundreds 4 ones

Solution:

(24x1000) + (7x100) + (4x1)

24000 + 700 + 4

24704

$$\begin{array}{r} 24000 \\ 700 \\ +\quad 4 \\ \hline 24704 \\ \hline \end{array}$$

Activity

Read and write the following in figures.

1. Five hundred thousand , two hundred seventy.
2. Twenty three thousand two.
3. One thousand , two hundred forty nine.
4. Two thousand , twenty four.
5. Nine hundred ninety thousand nine.
6. 251 thousands 345 units.
7. 7 thousand 2 hundred 7 ones.
8. Two thousand one.

Forming numerals from digits.

Examples

1. Write the smallest and the largest number that can be formed through the digits below 4, 5 and 2.

Soln

Note: * To find the smallest number formed just arrange the given digits into ascending order (from the smallest to the biggest).

*** To find the largest number formed, just arrange the given digits into descending order (from biggest to the smallest).**

4, 5, 2

Smallest number formed = 245

Largest number formed = 542

2. Form the smallest and the largest three digit numbers that can be formed from 4, 0, and 3.

Soln

Smallest number formed = 304

Largest number formed = 430

Activity

1. Form the smallest and the largest number that can be formed from the digits below;

a). 4, 6, 7 ______

b). 1, 4, 2 ______

c). 6, 0, 9 ______

d). 2, 6, 9 ______

2. Form all the possible three digit numbers that can be formed from 6, 7 and 8.

More about forming Numerals from digits.

Examples

Given digits 2, 6 and 7;

a). Form the smallest and the largest three digit numbers that can be formed.

Smallest number formed = 267

Largest number formed = 762

b). Find the sum between the largest and the lowest number formed.

Soln

$$\begin{array}{r} 762 \\ +267 \\ \hline 1029 \\ \hline \end{array}$$

c). What is the difference between the smallest and the largest number formed.

$$\begin{array}{r} 762 \\ -267 \\ \hline 495 \\ \hline \end{array}$$

Activity

1. Given digits 5, 6, 3. Use the digits given to;

a). Form the possible three digit numbers.

b). Find the smallest and the largest number formed.

c). Work out the difference between the smallest and the largest number formed.

d). Find the sum of the largest and the smallest number formed.

MORE TASKS

1. Given Digits 2, 4, 1, 0.

a) Form the smallest numeral from digits.

b) Form the biggest numeral from the digits.

c) Workout the sum of the two numerals formed.

d) Write the largest number in words.

e) Expand the least numeral in value form.

f) Calculate the range between the two numerals formed.

g) Draw an abacus and show the range above.

Expanding whole numbers up to 999, 99.

The method of rewriting a number using value of digits is the expanded form.

Whole numbers can be expanded in 3 ways i.e;

- Values
- Using place values
- Using powers of ten / exponents / indices.

Examples.

1. Expand as instructed.

927 (Using values)

$(9x100) + (2x10) + (7 x 1)$

$900 + 20 + 7$

2. 8430 (Using place values)

$(8x1000) + (4x100) + (3x10) + (0 x 1)$

3. 34678 (Using powers of ten)

$(3x10000) + (4 x 1000) + (6x100) + (7 x 10) + (8 x 1)$

$(3x10x10x10x10) + (4x10x10x10) + (6 x 10 x 10) + (7 x 10) + (8x1)$

$(3x10^4) + (4 x 10^3) + (6x10^2) + (7x10^1) + (8 x 10^0)$

Activity

Expand the following using values.

1. 34678

2. 109003

3. 5505

4. 3 4 7

Write these in expanded form using place values.

i) 3209

ii) 894

iii) 5469

Expanding whole numbers as multiples of ten.

Example 1: Expand 13456.

13456 =10000+3000+400+50+6

= (1x10x10x10x10) + (3x10x10x10)

(4x10x10) + (5x10) + (6x1)

Example 2: Expand 785.

785 = 700+80+5

= (7x10x10) + (8x10) + (5x1)

Activity

Expand the following using multiples of ten.

1. 4854

2. 17489

3. 4577

4. 5962

Writing expanded numbers in short form.

Example.

1. What number has been expanded to give;

2000 + 400 + 30 + 2

```
  2000
   400
    30
+    2
  2432
```

2. What number has been expanded to give;

2000 + 400 + 30 + 2

```
  2000
  0400
  0030
 +0002
  2432
```

3. Write these in short

(7x10,000) + (2 x 100) + (3x10)

Solution:

70,000 + 200 + 30

```
  70000
    200
+    30
  70230
```

Activity

Write the following in short.

1. (3x1000) + (5x100) + (1x10) + (3 x 1)

2. 400000 + 70000 + 500 + 90 + 8

3. (8x100000) + (8x100) + (3x10) + (5x1)

4. (9x1000) + (7x100) + (8x1)

5. (4 x 1000) + (8x100) + (7x1)

6. (8x10000) + (9x1000) + (9x100) + (9 x 10)

More about expanded numbers.

Find the expanded numeral below.

a) $(4\text{x}10^3) + (8\text{x}10^2) + (7\text{x}10^1) + (1\text{x}10^0)$

(4x10x10x10)+(8x10x10)+(7x10)+(1x1)

4000 + 800 + 70 + 1

4871

b) $(7\text{x}10^5) + (6\text{x}10^2) + (7\text{x}10^0)$

(7x10x10x10x10x10)+(6x10x10)+(7x1)

(7x100000) + (6x100) + 7

700000 + 600 + 7

700607

Activity

Find the expanded numeral in each of the following.

1. $(5 \times 10^4) + (6 \times 10^2) + (6 \times 10^0)$

2. $(9 \times 10^1) + (8 \times 10^0)$

3. $(3 \times 10^5)+(7 \times 10^4)+(6 \times 10^2)+(9 \times 10^0)$

4. $(1 \times 10^3)+(4 \times 10^2)+(2 \times 10^1)+(7 \times 10^0)$

Place values of decimals.

Example 1. Write the place values of the following digits below.

a) 0.75

O	Ths	Hths
0	. 7	5

Hundredths

Tenths

Ones

b) 24.312

T	O	Tths	Hths	Thths
2	4	3	1	2

Thousandths

Hundredths

Tenths

Ones

Tens

Activity

Write the place values of the following digits below.

1. 0.2

2. 0.13

3. 2.6

4. 64.521

5. 0.75

Values of the digits in decimals.

1. 37.62

T	O	Tths	Hths
3	7	6	2

$\left(2 \times \frac{1}{100}\right) = 0.02$

$\left(6 \times \frac{1}{10}\right) = 0.6$

$(7 \times 1) = 7$

$(3 \times 10) = 30$

2. 0.84

O	Ths	Hths
0	. 8	4

$\left(4 \times \frac{1}{100}\right) = 0.04$

$\left(8 \times \frac{1}{10}\right) = 0.8$

$(0 \times 1) = 0$

Activity

Find the value of each digit.

1. 0.25

2. 9.5

3. 0.125

Writing decimals in words.

1. Write 0.75 in words.

O	Ths	Hths
0	. 7	5

$\frac{0.75}{100}$ - seventy five hundredths

0.75 - seventy-five hundredths

2. 450.65

H	T	O	Tths	Hths
4	5	0	6	5

450 and $\frac{65}{100}$

Four hundred fifty and sixty-five hundredths.

Activity

Write the following decimal fractions in words.

1. 0.2

2. 0.75

3. 0.48

4. 12.8

4. 1.5

5. 0.62

Expanding wholes and decimals.

Example

1. 8.25

   8.25 = 8 + 0.2 + 0.05

2. 78.85 = 70 + 8 + 0.8 + 0.05

3. 364.27 = 300 + 60 + 4 + 0.2 + 0.07

Activity

Expand the following.

1) 13.84

2) 68.03

3) 135.28

4. 98.75

5. 60.78

Rounding off whole numbers up to the nearest 10,000

If a digit on the right of the required place value is less than 5 i.e 0, 1, 2, 3, 4 , you are not required to round off.

And if a digit on the right of the required place value is 5 or greater than 5 i.e 5,7,8,9, you round off. The digit in the required place value changes by adding 1 to it.

The digits on its right changes to zeros (0).

Examples.

Round off 5962.

a) nearest tens

5962

required place value (tens)

5960

$$\begin{array}{r} 5960 \\ +\ \ 0 \\ \hline 5960 \\ \hline \end{array}$$

(round down since 2 is less than 5)

$5962 \simeq 5960$

b) nearest hundreds.

5962

Required place value (hundreds)

5900

(round up since 6 is greater than 5)

$$\begin{array}{r} 5900 \\ +\ 1 \\ \hline 6000 \\ \hline \end{array}$$

$5962 \simeq 6000$

2. Round off 4365 to the nearest thousands.

5962

Required place value (thousands)

0 (round down since 3 is less than 5)

$$\begin{array}{r} 4000 \\ +0 \\ \hline 4000 \end{array} \quad 4365 \simeq 4000$$

3. Round off 174,898 to the nearest ten thousands.

174 898

(Required place value(tenthousands)

0 (round down since 4 is less than 5.)

$$\begin{array}{r} 170,000 \\ +0 \\ \hline 170000 \end{array} \quad 174\,898 \simeq 170000$$

Note:

The symbol for rounding off numbers is $\simeq$

Activity

Round off the following as instructed.

a) Hundreds
4564

b) thousands
4564

c) tens
762

d) tens
235

e) Hundreds
4854

f) hundreds
702

2. Round off 4365 to the nearest thousands.

5962

Required place value (thousands)

0 (round down since 3 is less than 5)

$$\begin{array}{r} 4000 \\ +0 \\ \hline 4000 \\ \hline \end{array}$$

4365 ≏ 4000

3. Round off 174,898 to the nearest ten thousands.

174 898

(Required place value(tenthousands)

0 (round down since 4 is less than 5.)

$$\begin{array}{r} 170,000 \\ +0 \\ \hline 170000 \\ \hline \end{array}$$

174 898 ≏ 170000

Note:

The symbol for rounding off numbers is ≏

Activity

Round off the following as instructed.

a) Hundreds
4564

b) thousands
4564

c) tens
762

d) tens
235

e) Hundreds
4854

f) hundreds
702

g) Ten thousands
86347

TOPIC 3: OPERATIONS ON WHOLE NUMBERS.

Words related to addition.

- Sum
- Total
- Increase
- Altogether
- Perimeter
- Profit

Adding of whole numbers upto 999 999.

Addition is a process of getting the sum of two or more numbers.
The result of addition is the sum or amount.
If the sum of digits is greater than 9, regroup to the next place value.

Example:

M I 35489 + 53679

$$\begin{array}{r} 3\,5\,4\,8\,9 \\ +\ 5\,3\,6\,7\,9 \\ \hline 8\,9\,1\,6\,8 \\ \hline \end{array}$$

Add ones 9 + 9 = 18
Add tens 1+8 + 7 = 16
Add hundreds1+4+6=11
Add thousands 1 + 5 + 3 = 9
Add ten thousands 3 + 5 = 8

M II

$$\begin{array}{r} 35489 = 30{,}000 + 5000 + 400 + 80 + 9 \\ 53679 = 50{,}000 + 3000 + 600 + 70 + 9 \\ \hline 80{,}000 + 8000 + 1000 + 150 + 18 \\ \hline \end{array}$$

$$\begin{array}{r} 80{,}000 \\ 8000 \\ 1000 \\ 150 \\ 18 \\ \hline 89168 \\ \hline \end{array}$$

Evaluating facts on addition of whole numbers.

Examples:

1. What is the sum of 3416 and 4293?

soln

$$\begin{array}{r} 3\ 4\ 1\ 6 \\ +\ 4\ 2\ 9\ 3 \\ \hline 7\ 7\ 0\ 9 \\ \hline \end{array}$$

2. Namubiru got 49361 votes during her elections and Shafik got 83124 votes. If all the pupils voted, Find the number of votes they got altogether.

$$\begin{array}{r} 4\ 9\ 3\ 6\ 1 \\ +\ 8\ 3\ 1\ 2\ 4 \\ \hline 1\,3\,2\ 4\ 8\,5 \\ \hline \end{array}$$ Votes

Excercise

1. Find the sum of 9623 and 11212.

2. The head teacher of our School brought 144 cups at the beginning and the director brought 639 cups. How many cups did they bring altogether?

3. Add the following numbers.

a) 405 132
+ 152 307
____

b) 518 320
+ 136 414
____

c) 735 066
+ 190 499
____

4) 769 430 + 875 = __________

Subtraction of whole numbers

Words to mean subtraction.

Subtract, minus, range, difference etc.

Examples:

1. Subtract:
   96329
   \- 74216
   ____

96329
\- 74216
22113

subtract ones 9 - 6 = 3
subtract tens 2 - 1 = 1
subtract hundreds 3 - 2 = 1
subtract thousands 6 - 4 = 2
subtract ten thousands 9 - 7 = 2

2. Subtract:
   953062
   \- 375184
   577878

Activity

Subtract the following.

1) 134576
\- 604
____

2) 83679
\- 21429
____

3) 530108
\- 311209
____

4) 100 600
\- 6400
____

4) 100 600
\- 6400
____

Evaluating facts on subtraction of whole numbers

Examples

1. Find the difference between 4934 and 1602.
   4934
   \- 1602
   3332

2. What is the difference between 7632 and 1396
   7632
   \- 1396
   6236

Activity

1. Find the difference between 463 and 213.

2. What is the difference between 97634 and 16934.

3. Charles planted 2000 trees and 1764 did not germinate.
   How many trees are growing now on his farm?

4. Micheal reported with 2 dozens of books at school and 11 books were stolen. How many books did Micheal remain with?

Multiplication of whole numbers.

- Multiply
- of
- Times
- Product.

Examples:

1. Multiply: 43 x 3

MI- Using repeated addition

$$\begin{array}{r} 43 \\ +43 \\ \underline{43} \\ \underline{129} \end{array}$$

MII- Using direct multiplication

$$\begin{array}{r} 43 \\ \underline{x3} \\ \underline{129} \end{array}$$

Multiply 43 by ones
43 x 3 = 129

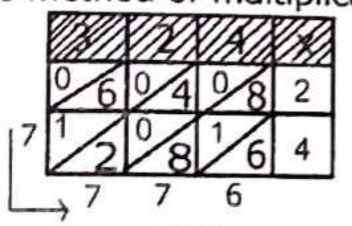
2. Multiply: 3 2 4

$$\begin{array}{r} 3\ 2\ 4 \\ \underline{x\ 2\ 4} \\ 1296 \\ \underline{+\ 6480} \\ \underline{7776} \end{array}$$

Multiply 324 by ones
324 x 4 = 1296
Multiply 324 by tens
324 x 20 = 6480

Using Latice method of multiplication

$$\begin{array}{r} 3\ 2\ 4 \\ \underline{x\ 2\ 4} \end{array}$$



7776

Activity

Workout the following.

1. 23 x 3
2. 250 x 4
3. 175 x 12
4. 764 x 32

6. Multiply. 3644

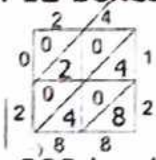
$$\begin{array}{r} 3644 \\ \underline{x\ 50} \end{array}$$

More about Multiplication of whole numbers.

Examples:

1. A box contains 24 books. How many books are in 12 boxes of books?

$$\begin{array}{r} 2\ 4 \\ \underline{x\ 1\ 2} \\ \underline{288} \end{array}$$



288 books

Activity

1. In P.5 class there are 56 pupils. If Mr. Opaade gave 4 pens to each pupil, How many pens did Mr. Opaade buy?

2. Find the product of 639 and 18.

3. How many eggs are on 100 trays if one tray contains 30 eggs?

Division of whole numbers.

Words to mean division.

- Share
- Divide
- Quotient

Examples.

1. Divide:

```
            062
        4 | 248 ——► (2÷4) impossible = 0
 (0x4) =   -0↓
            24  ——► (24÷4) = 6
 (6x4) =   -24↓
              8 ——► (8÷4) = 2
 (2x4) =      8
```

$248 \div 4 = 62$

2. What is the quotient of 2525 and 5?

```
             0505
         5 | 2525 ——► (2÷5) impossible = 0
 (0x5) =    -0↓
             25   ——► (25÷5) = 5
 (5x5) =     25↓
               2  ——► (2÷5) = 0
 (0x5) =       0↓
               25 ——► (25÷5) = 5
 (5x5) =       25
               ..
```

$2525 \div 5 = 505$

3. $3 \overline{)303}$

4. $6 \overline{)180}$

5. $12 \overline{)672}$

6. $3 \overline{)441}$

7. $6 \overline{)2716}$

Activity

1. 48 ÷ 2

2. 200 ÷ 4

More about division of whole numbers.

Examples.

1. Share 144 sweets among 2 boys.

$$
\begin{array}{r l}
 & \,072 \\
 & 2\overline{)144} \rightarrow (1 \div 2)\text{ impossible} = 0 \\
(0\times2) = & -0 \\
 & 14 \rightarrow (14 \div 2) = 7 \\
(7\times2) = & -14 \\
 & 4 \rightarrow (4 \div 2) = 2 \\
(2\times2) = & 4 \\
\end{array}
$$

Each boy got 72 sweets.

DIVISION OF BIGGER NUMBERS.

1. Divide $15\overline{)12075}$

$$
\begin{array}{r l}
 & \,803 \\
 & 15\overline{)12075} \\
(8\times15) = & -120 \\
 & 75 \\
(0\times15) = & -0 \\
 & 75 \\
(3\times15) = & -75 \\
 & -- \\
\end{array}
$$

2. Simplify: $7000 \div 200$

$$= \frac{70\cancel{00}}{2\cancel{00}}$$

$$= \frac{70}{2} = 35$$

Activity

1. $9685 \div 3$

2. $8340 \div 12$

3. $13332 \div 11$

4. $4000 \div 200$

5. $707 \div 7$

6. $5405 \div 23$

Activity

1. Namukose shared 36 books among her 6 children. How many books did each child get?

2. Share 144 by 3

3. Mr. Opio divided sh.1445 equally among his 15 children. How much did each child get?

4. A farmer planted 39 seedlings in 3 plots. How many were planted on each plot?

Divisibility test.

1. Any number is divisible by 2 if its digit is an even number.

2. Any number is divisible by 3 if its sum of digits is multiple of 3.

3. Any number is divisible by 4 if its last two digits form a number which is a multiple of 4.

4. Any number is divisible by 5 if its last digit is 5 or 0.

5. Any number is divisible by 6 if the sun of its digits is a multiple of 9.

Combined operation of Numbers (BODMAS)

BODMAS = **Brackets**
Of
Division
Multiplicateion
Addition
Subtraction

Examples

1. Workout
5 + (3x10)
5 + 30
35

2. Workout
2 - 8 + 9
(2 + 9) - 8
11 - 9 = 3

3. Workout
(8 + 9) x 10
15 x 10
= 150

4. Workout
5 x 12 ÷ 4
5 x (12 ÷ 4)
5 x 3 = 15

Note: Division and multiplication in the two operations in BODMAS don't change position. You work upon them there and then.
Only addition and subtraction change the positions and addition comes first.

Activity

Work out the following numbers.

1. $\frac{1}{2}$ of 10 + 15 ÷ 5

2. 6 - 10 + 7

3. 32 - 40 + 18

4. $6 \div 6 + 2 - 3$	6. $28 \times 2 - 1$
5. $(10 + 10) \div 5$	7. $(8 - 5) - (3 \times 2)$

STATISTICS

Terms used in statistics.

- Mode
- Range
- Median
- Modal frequency
- Mean / average

Mode: refers to the number which appears many times.

Examples:

1. Given 2, 3, 0, 6, 3 and 4 . Find

a) Mode

No	Frequency
0	1
2	1
3	2
4	1
6	1

Mode is **3** and modal frequency is **2**.

2. Mark did a test and scored the following results; 90, 40, 30, 90, 60, 90.

(i)Find his modal mark

Mode	Frequency
90	3
40	1
30	1
60	1

His modal mark is 90

(ii). Calculate his modal frequence

Note: Modal frequency is the number of times a number appears.

Modal frequency = 3

Activity

For the following data given, find the modal mark and modal frequency for each data.

a). 1. 3, 2, -3, 2, 4, 2, 6

b). -3, -1, -3, -1, -4, -1, -3.

c). 60, 40, 120, 60, 120, 120

Finding Range and Median.

Note: * Range is the difference between the highest and the lowest value of data given.

* Median is the number that comes in the middle after arranging the given data in ascending order.

Examples;

Given; 6, 7, 4, 2, 4, and 4.

a) Find Range

Range = Highest - Lowest value.

= 7 - 2

= 5

b). Find median.

2, 4, 4, 4, 6, 7

$\frac{4+4}{2}$ = 8/2 =4

Activity

Find the median and Range of the following data.

a) 70, 60, 80, 60, 40, 60

b). -1, 2, 0, -3, -1

c). 50, 51, 52, 73, 63, 83

d) 100, 140, 80, 93, 120

Calculating Mean (Average)

Mean is the Ratio between the sum of given data and the number of given data.

Note: Ratio means (division).

Formula for mean/average

$$A = \frac{\text{Sum of given data}}{\text{No. of given data}}$$

Examples

1. Find the average of 2, 4, 0, and 6

Soln

$$\text{Average} = \frac{\text{Sum of given data}}{\text{No. of given data}}$$

$$= \frac{2 + 4 + 0 + 6}{4}$$

= 12/4 = 3

2. Calculate the mean of 20, 12, 10, 8

$$\text{Average} = \frac{\text{Sum of given data}}{\text{No. of given data}}$$

$$= \frac{20 + 12 + 10 + 8}{4}$$

= 50/4 = 25/2 = $12\frac{1}{2}$

ctivity

. Calculate the average/mean of the ollowing given data.

. 70, 60, 90, and 80.

2. 16000, 18000, 7500, and 1300

3. 66°F, 74°F, 30°F, 82°F, 55°F, and 65 °F

4. 3, 6, 9, 12 and 15

Using the symbols >, < or =

Using means the symbols >, < or =.

> means greater than.

< means less than.

= means equal to.

Examples.

1. 2 + 3 ________ 2 x 3

5 ________ 6

5 ____<____ 6

∴ 2 + 3 ____<____ 2 x 3

2. ½ of 16 ________ ¼ of 36

$\frac{1}{\cancel{2}_1} \times \cancel{16}^{8}$ ________ $\frac{1}{\cancel{4}} \times \cancel{36}\,\cancel{18}\ 9$

8 ____<____ 9

∴ ½ of 16 ____<____ ¼ of 36

Activity

Compare the following using =, <, >

1. 10 x 10 ________ 10 x 2

2. 3 x 3 ________ 3 + 3

3. 2 x 3 ________ 2 + 3

4. 75 gm ________ 1kg

COUNTING IN BASE 5.

(Non-Decimal systems)

The system of counting in groups of five is Base five numeration system. Another name for base five is quinary system.

Other groupings include:

- Grouping in twos - base two (Binary)
- Grouping in threes - base three (Ternary)
- Grouping in fours - base four (quaternary)
- Grouping in fives - Quinary base.

Note:
In any base, the digits used must be less than the given base.

Decimal system is the grouping of numbers in tens (base ten)

Finding Place values in base five.

Examples.

1. Show the place value of each digit in the number 342_{five}.

3 4 2 five
- 2 — Ones
- 4 — Fives
- 3 — Five fives

2. Find the place value of each digit in 10234_{five}

1 0 2 3 4
- 4 — Ones
- 3 — Fives
- 2 — Five fives
- 0 — Five fives fives
- 1 — Five fives fives fives

Activity

Give the place values of each digits below.

1. 120_{five}

2. 312_{five}

3. 234_{five}

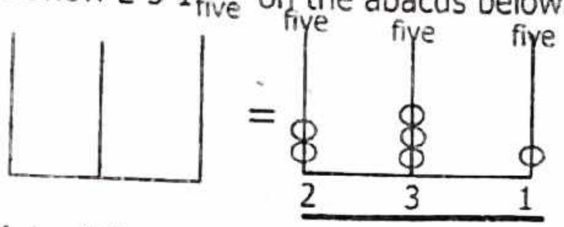
4. 1304_{five}

5. 1024_{five}

Representing base five numbers on the abacus.

Example:

1. Show 2 3 1_{five} on the abacus below.

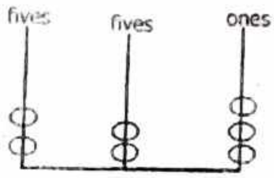


Note: When you don't apply place values on the abacus its marked wrong.

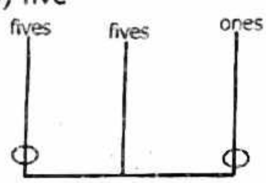
Activity

1. Write the numbers shown on the abacus

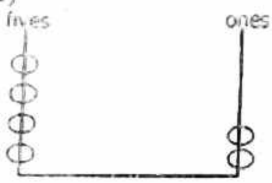
a) five



b) five



c)



2. Represent the number below on the abacus

a). 102 five

b). 144 five

Expanding In Base Five.

We can expand a number in base five and in base ten.

Examples

1.Expand 213 five in base five.

Five fives	fives	ones
2	1	3
5^2	5^1	5^0

$213_{five} = (2 \times 5^2)_{ten} + (1 \times 5^1)_{ten} + (3 \times 5^0)_{ten}$

Method II

$213_{five} = (2 \times 100)_{five} + (1 \times 100)_{five} + (2 \times 100)_{five}$

2.Expand 340 five in base five

$340_{five} = (3 \times 5^2)_{ten} + (1 \times 5^1)_{ten} + (2 \times 5^0)_{ten}$

Activity

Expand the following.

a) 34_{five}

b). 2034_{five}

e) 431_{five}

f) 210_{five}

g) 313_{five}

h) 32_{five}

I) 283_{five}

Finding Values in base five.

Examples.

Write the values of each digit in the following.

a) 213_{five}

$3 \times 1 = 3$

$1 \times 5 = 5$

$2 \times 5 \times 5 = 50$

b) 2034_{five}

$4 \times 1 = 4$

$3 \times 5 = 15$

$0 \times 5 \times 5 = 0$

$2 \times 5 \times 5 \times 5 = 250$

State the value of the underlined digit.

a) 24_{five}

$2 \times 5 = 10$

b) 243_{five}

$2 \times 5 \times 5 = 50$

Activity

Find the value of each digit in the following;

a) 113_{five}

b) 202_{five}

c. 143_{five}

Calculate the value of 2 in the following.

a) 2104_{five}

b) 342_{five}

c) 1203_{five}

d) 24_{five}

Writing base five numbers in words.

Examples.

1. Write 43_{five} in words

43_{five} = four, three, base five.

2. Write 213_{five} in words

213_{five} = Two, one, three, base five.

Activity

Write the following in words.

1. 142_{five}

2. 32_{five}

3. 241_{five}

4. 123_{five}

5. 41_{five}

Changing from base ten to base five.

To change base ten numbers to base five.

Keep on dividing by 5 while recording the remainders.

The answer is a number formed by remainders recorded from the last remainder.

Examples.

Change 23_{ten} to base five.

Method I.

Group 23 in fives.

23 = (IIIII) (IIIII) (IIIII) (IIIII) III

4_{fives} and 3 $_{ones}$ = 43_{five}.

Method II.

Base	Number	Remainder
5	23	3.
	4	

23_{ten} = 43 $_{five}$

Example 2.

Change 40 to base five.

B	N	R
5	40	0
5	8	3
	1	

130 five

Activity

Change the following to base five.

a) 9 ten

b) 32 ten

c) 42ten

d) 72ten

e) 14ten

f) 55ten

g) 26ten

h) 101ten

Changing from base five to base ten.

To change base five numbers to base ten.

- Expand the given number using base ten values.

Examples.

Convert 144_{five} to base ten.

144_{five}

$4 \times 1 = 4$

$4 \times 5 = 20$

$1 \times 5 \times 5 = +25$

49

$144_{five} = 49_{ten}$

2) Change 123_{five} to base ten.

123_{five}

$3 \times 1 = 3$

$2 \times 5 = 10$

$1 \times 5 \times 5 = 25$

38ten

Note: Base ten can also be called standard base, decimal base, unit base, mother base, binary base, invisible base, every day base, etc.

Activity

Change the following to base ten.

a) 21_{five}

b) 201_{five}

c) 112_{five}

d) 421_{five}

e) 34_{five}

f) 212_{five}

What base ten number is equivalent to

a) 111_{five}

b) 104_{five}

c) 1012_{five}

Adding numbers in base five.

Examples: 1. Add $102_{five} + 111_{five}$

soln

$$\begin{array}{r} 1\ 0\ 2_{five} \\ +\ 1\ 1\ 1_{five} \\ \hline 2\ 1\ 3_{five} \\ \hline \end{array}$$

2. Add a) $4_{five} + 3_{five}$

$$\begin{array}{r} 4_{five} \\ +\ 3_{five} \\ \hline 12_{five} \\ \hline \end{array} \quad 7 \div 5 = 1r2$$

Note: Adding in base five is done the same way as it is done in base ten. If the sum of digits is equal or greater than five, regroup to the next place value.

Activity

Work out the following.

a) $2_{five} + 4_{five}$

b) $23_{five} + 40_{five}$

c) $3_{five} + 12_{five}$

d) $43_{five} + 234_{five}$

e)
$$\begin{array}{r} 1\,0\,2_{five} \\ +\,2\,3\,4_{five} \\ \hline \\ \hline \end{array}$$

f)
$$\begin{array}{r} 1\,4\,2_{five} \\ +\,2\,1\,3_{five} \\ \hline \\ \hline \end{array}$$

g)
$$\begin{array}{r} 1\,2\,3\,0_{five} \\ +\,1\,1\,4\,2_{five} \\ \hline \\ \hline \end{array}$$

Subtracting numbers in base five.

Examples.

Subtract: $44_{five} - 21_{five}$

$$\begin{array}{r} 44_{five} \\ -\,21_{five} \\ \hline 23_{five} \end{array}$$

b) $41_{five} - 13_{five}$

$$\begin{array}{r} 4\,1_{five} \\ -\,1\,3_{five} \\ \hline 2\,3_{five} \\ \hline \end{array}$$

1+5 =6 / 6 - 3=

Note: In subtraction of base five, in case of borrowing we borrow the bas (5) from the next place value.

Activity

Work out the following.

1. $32_{five} - 22_{five}$

2. $44_{five} - 31_{five}$

3. $211_{five} - 44_{five}$

4.
$$\begin{array}{r} 4\,3_{five} \\ 1\,4_{five} \\ \hline \\ \hline \end{array}$$

5.
$$\begin{array}{r} 1\,3\,2_{five} \\ -\,4\,3_{five} \\ \hline \\ \hline \end{array}$$

6. 310_{five}
$- \ 20_{five}$

7. 222_{five}
$- \ 114_{five}$

Multiplication of numbers in the non-decimal system.

1. 203_{five}
$\times \ 2_{five}$
411_{five}

$6 \div 5 = 1 r 1$
$(0 \times 2) = 0+1$
$(2 \times 2) = 4$

2. 304_{five}
$\times \ 2_{five}$
1113_{five}

$8 \div 5 = 1 r 3$
$(0 \times 2)+1 = 1$
$2 \times 3 = 6$
$6 \div 5 = 1r1$

3. 501_{seven}
$\times \ 4_{seven}$
2604_{seven}

$1 \times 4 = 4$
$3 \times 4 = 12 \div 7$
$= 1r5$
$20 \div 7 = 2 r 6$
$1 \times 4 = 4$
$0 \times 4 = 0$
$5 \times 4 = 20 \div 7 = 2r6$

Activity

1. 32_{five}
$\times \ 4_{five}$

2. 102_{five}
$\times \ 2_{five}$

3. 114_{five}
$\times \ 4_{five}$

4. 234_{eight}
Activity 2_{eight}

5. 234_{five}
$\times \ 2_{five}$

Clock arithmetic.

Example 1: Write 3 in finite 5.
3 = 3 (finite 5)

Example 2: Change 25 to finite 7
$25 \div 7 = 3 r 4$
so 25 = 4 (finite 7)

Activity

Express in finite 5 or finite 7.

1. 2 finite 5

2. 11 finite 5.

3. 18 finite 5

4. 18 finite 5

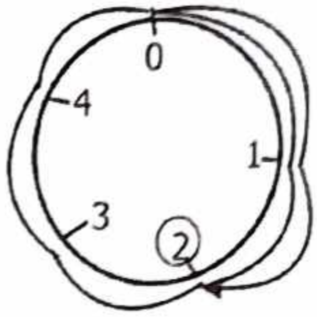
5. 10 finite 7

6. 20 finite 7

6. 12 finite 7

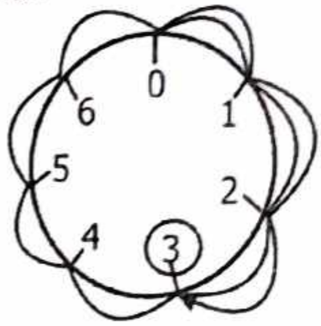
Addition in finite 5 and using a dial method.

1. Add: 4 + 3 = ____ (finite 5) using dial.



4 + 3 = 2 finite 5

2. Add: 4 + 3 = ____ (finite 7) using dial.



4 + 3 = 3 finite 7

Activity

Use a dial to workout.

1. 2 + 2 = __________ (finite 5)

2. 4 + 5 = ______(finite 7)

5. 4 + 4 = ______ (finite 5)

3. 2 + 2 + 3 = ______(finite 5)

4. 3 + 1 + 3 =______(finite 7)

TOPIC 4:

NUMBER PATTERNS AND SEQUENCES

Increasing and decreasing progression.

A set of numbers that follow a certain pattern form or sequence.

A sequence is an ordered list of numbers.

Each number in the sequence is called a term.

The arithmetic progression is obtained by adding the same number to the previous term or subtracting the same term from the previous term. (number)

Example 1:

Find the next three terms in the progression.

0, 2, 4, 6, 8, 10, 12

+2 +2 +2 +2 +2 +2

Keep adding 2.

Example 2:

Find the next 2 terms in the sequence.

87, 81, 75, 69, 63, 57

-6 -6 -6 -6 -6

Keep subtracting 6..

Activity

Find the next two terms in;

1. 2, 9, 16, 23, 30, _______, _______

2. 4, 14, 24, 34, _______, _______

3. 320, 360, _____, 400, ______, ______, 520

4. 12, 21, 30, _______, _______

5. 16, 14, 12, _______, _______, 6, 4

6. 27, 22, 24, _______, _______

7. 150, _______, 100, _______, 75, _______, _______, 25

8. 900, _______, 700, 600, _______, _______, 300, _______, 100.

TYPES OF NUMBERS

- Even numbers.
- Odd numbers.
- Triangular numbers.
- Rectangular numbers.
- Square numbers.
- Prime and composite numbers.

Even numbers and Odd numbers.

Numbers which are divisible by 2 are called even numbers.
Numbers which leave a remainder 1 after dividing by 2 are odd numbers.
A set of odd numbers = 1, 3, 5, 7, 9, 11, 13, ...
A set of even numbers = 0, 2, 4, 6, 8, 10, 12, 14, ...
The last digit in all even numbers in ones place values are even.
The last digit in all odd numbers is ones place values are odd.

Example 1.

List all even numbers between 20 and 30.
= { 22, 24, 26, 28}
List all odd numbers between 15 and 26 ,
= {17, 19, 21, 23, 25}

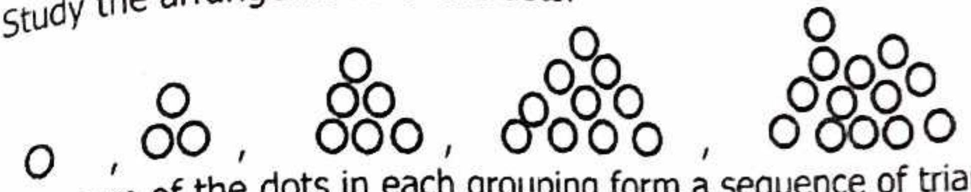
Activity

1. List all even numbers less than 20.
....
2. List all odd numbers less than 15.
....
3. What do you understand by
   a) Even numbers
   ....
   ....
   b) Odd numbers
   ....
   ....
4. Say true or false.
   All numbers greater than 1 are even numbers.
   - is an even number.
   - is not an even number.
5. Write a set of the first seven even numbers. ....

6. Calculate the sum of the first 5 odd numbers.

Triangular patterns and triangle numbers.

Study the arrangements of the dots.



The sum of the dots in each grouping form a sequence of triangular number.

Triangular members form triangular patterns.
A set of triangular members = { 1, 3, 6, 10, 15,........}

Example 1.
Find the first 4 consecutive triangular numbers.

Solution.

1 - 1
1 + 2 - 3
1 + 2 + 3 - 6
1 + 2 + 3 + 4 - 10
{ 1, 3, 6, 10 }

Example 2.
Find the 5^{th} triangular number.

$\frac{1}{2}n(n+1) \qquad \frac{1}{2} \times 5\,(5+1)$

$\frac{1}{2} \times 5 \times \overset{3}{\cancel{6}} = 5 \times 3 = 15$

Activity

1. List all triangular numbers less than 20.

2. Write a set of triangular numbers between 10 and 20.

3. What is the 4th triangular number?

4. Find the next two numbers in 3, 6, 10, 15, ______, ______

5. Draw patterns using dots for the first 6 triangular numbers.

Rectangular patterns and rectangle numbers.

The rule for finding the next rectangular number is to multiply two consecutive counting numbers.

Rectangle number is twice a triangle number.

1 + 1	2
1 + 2 + 1	4
1 + 2 + 2 + 1	6
1 + 2 + 3 + 3 + 2 + 1	12

Examples.

Find the first five rectangular numbers.

2, 2 x 3 = 6, 3 x 4 = 12, 4 x 5 = 20, 5 x 6 = 30

2, 6, 12, 20, 30

Activity

1. Fill in the missing : 2, 6, 12, 20, ______, ______

2. What is the 5th rectangular number?

3. List the rectangular numbers between 10 and 30.

4. Write **true** or **false**.

a) A rectangular number is a number that forms a rectangle.

b) All rectangle numbers are triangular numbers.

c) Twice a triangular number results to rectangular number.

Square numbers and square patterns.

A square number is formed by multiplying a counting number by itself.

or

Adding two consecutive triangular numbers.

or

Adding consecutive odd numbers.

Example 1.

1 x 1 = 1 3 x 3 = 9

2 x 2 = 4 4 x 4 = 16........

1 1 + 3 = 4 3 + 6 = 9 6 + 10 = 16....................

1 1 + 3 = 4 1 + 3 + 5 = 9 ....

Note: A number that is multiplied to get a square number is called a square root.

Example 2.

a) What are the first 15 square numbers?

1 x 1 = 1
2 x 2 = 4
3 x 3 = 9
4 x 4 = 16
5 x 5 = 25
6 x 6 = 36
7 x 7 = 49
8 x 8 = 64
9 x 9 = 81
10 x 10 = 100
11 x 11 = 121
12 x 12 = 144
13 x 13 = 169
14 x 14 = 196
15 x 15 = 225

{1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, 225}

b) What is the 8^{th} square root?

= 8 x 8 = 64

Finding the square of numbers.

a) $7^2 = 7 \times 7$
$= 49$

b) 12
$12^2 = 12 \times 12$
$= 144$

c) 15
$15^2 = 15 \times 15$
$= 225$

d) 10
$10^2 = 10 \times 10$
$= 100$

Activity

1) Find the square of 3.

2. Find the value of x^2 if $x = 2$.

3. Simplify 9cm x 9cm.

4) What is the square of 9?

5. Workout: 7km x 7km.

6. Given that m=16, find the value of m^2.

Finding the square root.

1. 16

$\sqrt{16}$ =

2	16
2	8
2	4
2	2
	1

$\sqrt{(\cancel{2}\times2)\times(\cancel{2}\times2)}$

$\sqrt{16} = 2\times2$

$= 4$

2) 25

25

5 5

5 1

$\sqrt{(5\times5)}$

$\sqrt{25} = 5$

Activity

1. Solve $k^2 = 25$

2. Find the square root of the following.

a) 81

b) 49

d) 4

c) 121

e) 144

Activity

1. State true or false.

a) The product of a number by itself is a square root.

....

b) Some square numbers are also rectangular numbers.

....

c) The sum of triangular numbers is a square number.

....

d) All triangle numbers are square numbers.

....

2. Find the square of the following.

(a) 8

(b) 10

c) 11

d) 14

e) 7

Prime and composite numbers.

A prime number is a number with only two factors, 1 and itself.
e.g {2, 3, 5, 7, 11, 13, 17,................}

Composite numbers are numbers with more than two factors e.g
4, 6, 8, 9, 10, ...

Note: 1 is not prime number since it has only one divisor i.e Itself.

Examples.

List all prime numbers less than 10.

1, 2, 3, 4, 5, 6, 7, 8, 9, 10

= 2, 3, 5, 7

List all composite numbers between 10 and 20.

11, 12, 13, 14, 15, 16, 17, 18, 19

= 11, 13, 17, 19

Activity

a) List the prime numbers between;

i) 3 and 40

ii) 50 and 60

iii) 70 and 80

iv) 10 and 30

b) List the composite numbers between 1 and 20.

c) List the composite numbers between 20 and 50.

Finding factors of numbers

A factor of a number is a number that divides that number completely without a remainder.

Examples

Find the factors of 6.

soln

MI

$$\left.\begin{aligned} &= 6 \div 1 = 6 \\ &= 6 \div 2 = 3 \\ &= 6 \div 3 = 2 \\ &= 6 \div 6 = 1 \end{aligned}\right\} \text{These are all factors of 6}$$

F6 = {1, 2, 3, 6}

2. Find the factors of 24.

MII

F24 = 24 X 1 = 24
12 X 2 = 24
6 X 4 = 24
8 X 3 = 24

These are all factors of 24

F24 = {1, 2, 3, 4, 6, 8, 12, 24}

Activity

Find the factors of the following numbers.

1. 15
2. 6
3. 16
4. 81
5. 100
6. 144
7. 26
8. 14

Finding common factors of 2 numbers.

Examples.

Find the common factors of 6 and 4.

F6 = {1, 2, 3, 6}

F4 = {1, 2, 4}

Common Factors of 6 and 4 = { 1, 2}

Activity

Find the common factors of the following numbers.

1. 24 and 32
2. 12 and 18
3. 6 and 9
4. 40 and 30
5. 33 and 11

Calculating the Greatest Common Factor of numbers (G.C.F/H.C.F).

Examples.

1. Find the HCF of 12 and 8.

F12 = {1, 2, 3, 4, 6, 12}

F18 = {1, 2, 3, 6, 9, 18}

Common factors = {1, 2, 3, 6}

The HCF of 12 and 18 is 6.

2. Find the HCF of 4 and 12.

F4 = {1, 2, 3, 4}

F12 = {1, 2, 3, 6, 12}

Common factors = {1, 2, 4}

The HCF of 4 and 12 is 4.

Activity

Find the H.C.F/G.C.F of the following numbers;

1. 6 and 9
2. 12 and 18
3. 12 and 14
4. 32 and 28

Prime Factorization.

To find prime factors, divide the number starting with the smallest prime number until the quotient is 1.

We can find/get the prime factors in;

- Multiplication form.
- Subscript / set notation form.

Example 1.

Prime factorise 60. (Using factor tree)

60 **MI**

2 30

2 15

3 5

5 1

2 x 2 x 3 x 5 (multiplication)

2^1, 2^2, 3^1, 5^1 (subscript form)

MII **Using ladders method.**

2	60
2	30
3	15
5	5
	1

2x2x3x5

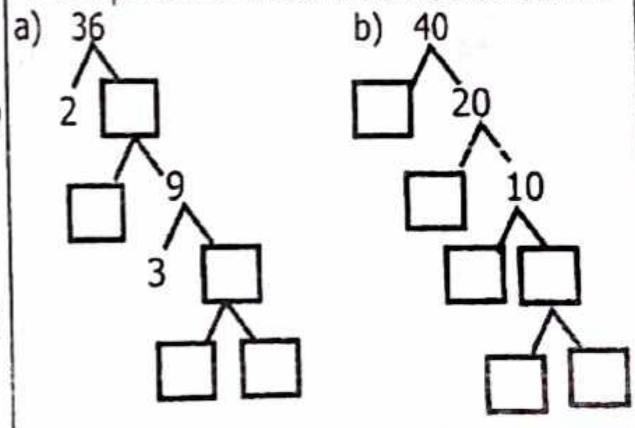
Activity

1. Prime factorise 12 and 5.

2. Find the prime factors of the numbers below using (multiplication form)
   i) 18
   ii) 42
   iii) 36
   iv) 90
   v) 72

3. Use the ladder method to prime factorize
   i) 60 ii) 40 iii) 36

4. Complete the factor tree method below.



Representing prime factors on venn diagrams.

Examples

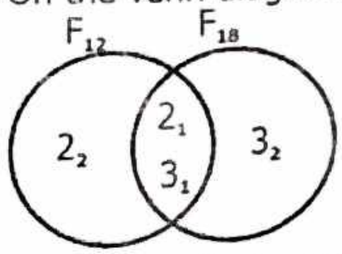
1. Represent the prime factors of 12 and 18 on the venn diagram.

Soln

pf 12 = $\begin{array}{r|l} 2 & 12 \\ \hline 2 & 6 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$ pf 18 = $\begin{array}{r|l} 2 & 13 \\ \hline 3 & 9 \\ \hline 3 & 3 \\ \hline & 1 \end{array}$

pf 12 = $\{2_1, 2_2\ 3_2\}$ & pf 18 = $\{2_1, 3_1\ 3_2\}$

On the venn diagram



Activity

Represent the prime factors of the following numbers on Venn diagrams.

1. 26 and 28

2. 38 and 54

3. 31 and 37

Finding the prime factorised number.

Multiply a set of prime factors to get the original factorised number.

Example:

Find the prime factorised number.

$2 \times 3 \times 3 \times 5 = 6 \times 15 = \mathbf{90}$

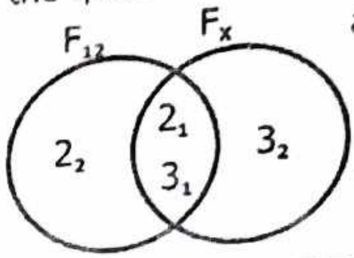
Find the number whose prime factorisation is;

$X = \{2^1, 3^1, 5^1\}$

$X = 2 \times 3 \times 5$

$X = 30$

Use the venn diagram below to answer the questions that follow.



a) Find the value of x.

$X = 2 \times 3 \times 3$

$X = 18$

b) Find GCF

2×3

$= 6$

c) Find the L.C.M

$2 \times 2 \times 3 \times 3$

$4 \times 9 = 36$

Activity

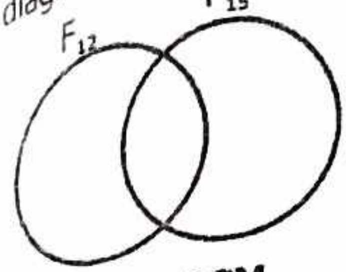
1. Which numbers have been prime factorised to get.

a) $2 \times 3 \times 5$ b) $2 \times 3 \times 7$

c) $2 \times 3 \times 11$ d) $2 \times 3 \times 5 \times 5$

e) 2×5 f) $3 \times 3 \times 1$

2. Represent the prime factors of numbers 12 and 15 on the venn diagram below.



b) Find the G.C.F and L.C.M.

GCF AND LCM

To find the GCF, we use;

- Listing method.
- Prime factors.
- List the factors of each number.
- Find the common factors to both numbers.
- 1 is always the lowest common factor (LCF).

Example 1.

Find the GCF of 12 and 18 (By listing).

$F_{12} = \{1, 2, 3, 4, 6, 12\}$

$F_{18} = \{1, 2, 3, 6, 9, 12\}$

Common factors = 1, 2, 3, 6

The GCF is 6.

Example 2.

Find the GCF of 30 and 36. (By using prime factors).

2	30
3	15
5	5
	1

2	36
2	18
2	9
3	3
	1

Common factors

{2, 3,}

2 x 3 = 6

Note: Using prime factors, first prime factorise the given numbers.

- Identify the common members.
- Find the product of the common factors to get the GCF.

Activity

List all the factors of each number.

ai) 6 ii) 28 iii) 45

bi) 10 ii) 36

ci) 15 ii) 12

di) 48 ii) 30

2. By using prime factorisation method, find the GCF of;

a) 12 and 15.

b) 18 and 30.

c) 21 and 35.

3. Find the greatest number that can divide 12 , 24 and 60 without a remainder.

Finding multiples of numbers.

A multiple is the product of a given number and another non- zero whole numbers.

Examples.

Find the multiples of 3 less than 20.

$1 \times 3 = 3$ // $3 \times 3 = 9$ // $5 \times 3 = 15$

$2 \times 3 = 6$ // $4 \times 3 = 12$ // $6 \times 3 = 18$

$M_3 = \{3, 6, 9, 12, 15, 18\}$

List the multiples of 6 between 20 and 40.

$M_6 = \{24, 30, 36\}$

Activity

1. Find the multiples of each of the following.

a) multiples of 5 less than 30.

b) multiples of 7 less than 100.

c) multiples of 5 less than 30.

2. Circle multiples of 5.

17, 10, 23, 12, 16, 24, 25

3. Write **true** or **false**.

a) 20 is a multiple of 6.

b) 6 is a multiple of 3.

c) 36 is a factor of 6.

4. Give the multiples of 4 between 10 and 50.

Finding Lowest common multiples (LCM).

- Lcm can also be called Least common multiple.

To find Lcm,
- List the multiples of each number.
- Choose the common multiples and select the least.

Example:

Find the Lcm of 4 and 6.
- Multiples of 4 = 4, 8, 12, 16, 20, 24, 28.....
- Multiples of 6 = 6, 12, 18, 24, 30, 36

Common multiples = {12, 24}

The Lcm of 4 and 6 is 12. Since it is the least.

2. Find the least number of sweets that can be shared by 8 and 18 children leaving no remainder.

Lcm of 8 and 18.

$M_8 = \{ 8, 16, 24, 32, 40, 48, 54 ...\}$

$M_{18} = \{ 18, 36, 54, 72, ...\}$

Lcm of 8 and 18 is 54.

Activity

1. Find the Lcm of the following;

a) 12 and 4

c) 18 and 28

b) 18 and 21

d) 7 and 14

e) 8 and 10

f) 8, 12 and 4

3. Calculate the least number that can be shared by 4 and 9 leaving no remainder.

TOPIC 5: FRACTIONS

Fraction is a part of a whole number.

Types of fractions.

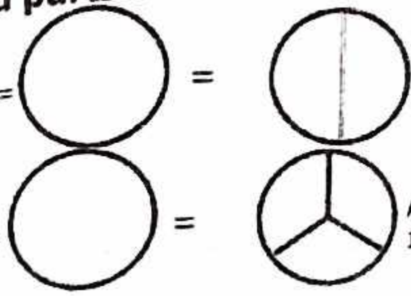
- Proper fractions
- Improper fractions
- Mixed numbers/ fractions
- Equivalent fractions
- Decimal fractions.

Names and parts of fractions.

Examples

1 whole 2/2 =

1 whole 3/3 =



A half 1/2

A third 1/3

Expressing improper fractions as mixed fractions.

Note: Improper fraction is a type of fraction where the numerator is greater than the denominator ie $\frac{13}{5}$

* A mixed fraction is a type of fraction which consists of a whole number and a proper fraction ie $2\frac{1}{3}$, $5\frac{1}{2}$, $3\frac{1}{2}$ etc.

Examples

1. Change $\frac{9}{5}$ as a mixed fraction.

soln

$$\begin{array}{r|l} & 1 \\ 5 & 9 \quad 9 \div 5 \\ (1 \times 5) & \underline{5} \\ & 4 \text{ (Remainder)} \end{array}$$

1 is the whole number.
4 is the numerator.
5 is the denominator.

$$\frac{9}{5} = 1\frac{4}{5}$$

2. Express $\frac{30}{7}$ as a mixed fraction

$$\begin{array}{r|l} & 04 \\ 7 & 30 \quad (3 \div 7) \text{ impossible} \\ (0 \times 7) & -\underline{0} \\ & 30 \quad (30 \div 7) = 4 \\ (4 \times 7) & -\underline{28} \\ & 2 \text{(Remainder)} \end{array}$$

$$\frac{30}{7} = 4\frac{2}{7}$$

Activity

Express the following improper fractions as a mixed fractions.

1. $\frac{15}{7}$
2. $\frac{37}{8}$
3. $\frac{44}{5}$
4. $\frac{50}{8}$
5. $\frac{23}{10}$

Expressing Mixed fractions as improper fractions.

Examples

1. Change $4\frac{2}{3}$ as an improper fraction.

soln Whole — $4\frac{2}{3}$ — Numerator, Denominator

$$= \frac{(D \times W) + N}{D}$$

$$= \frac{(3 \times 4) + 2}{3} = \frac{12 + 2}{3}$$

$$= \frac{14}{3}$$

2. Express $5\frac{1}{4}$ as an improper fraction.

$$5\frac{1}{4} = \frac{(D \times W) + N}{D} = \frac{(4 \times 5) + 1}{4}$$

$$\frac{20 + 1}{4} = \frac{21}{4}$$

Activity

Express the following fractions as an improper fractions.

1. $5\frac{1}{12}$
2. $4\frac{3}{8}$
3. $3\frac{1}{7}$
4. $1\frac{1}{18}$
5. $2\frac{2}{11}$
6. $5\frac{7}{11}$

Equivalent fractions.

Fractions that are equal in size and value are equivalent fractions.

Examples.

Find the next 3 equivalent fractions to the following fractions.

1. $\frac{1}{3} = \frac{1x2}{3x2} = \frac{2}{6}, \quad \frac{1x3}{3x3} = \frac{3}{9}, \quad \frac{1x4}{3x4} = \frac{4}{12}$
2. $\frac{6}{9} = \frac{6x2}{9x2} = \frac{12}{18}, \quad \frac{6x3}{9x3} = \frac{18}{27}, \quad \frac{6x4}{9x4} = \frac{24}{36}$

Activity

Find the equivalent fractions of the following.

1. $\frac{5}{10}$,
2. $\frac{2}{3}$
3. $\frac{3}{4}$
4. $\frac{4}{9}$,
5. $\frac{5}{8}$

Find the missing numbers.

a) $\frac{2}{3} = \frac{\square}{27}$

b) $\frac{4}{9} = \frac{\square}{18}$

c) $\frac{3}{4} = \frac{9}{\square}$

d) $\frac{4}{5} = \frac{12}{\square}$

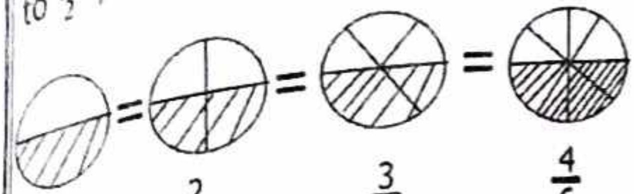
Note:
In order to obtain equivalent fractions, you multiply the original fraction by counting numbers starting from 2, to get the next equivalent fraction.

Finding equivalent fractions using diagrams.

Fractions that are equal in size and value are equivalent fractions.

Examples.

Find the next 3 equivalent fractions to $\frac{1}{2}$, using diagrams.



$\frac{1}{2}$ $\frac{2}{4}$ $\frac{3}{6}$ $=\frac{4}{6}$

Equivalent fractions upto

$\frac{1}{2}$, $\frac{2}{4}$, $\frac{3}{6}$, $\frac{4}{6}$

Activity

Use diagrams to find the following equivalent fraction; (3 on each)

1. $\frac{1}{3}$

2. $\frac{2}{3}$

3. $\frac{1}{10}$

4. $\frac{1}{4}$

5. $\frac{1}{6}$

Reducing fractions.

Examples.

1. Reduce $\frac{12}{24}$ using G.C.F.

Soln

The GCF of 12 and 24 is 12.

$\frac{12}{24}$ reduce by 12

$\frac{12 \div 12}{24 \div 12}$

$= \frac{1}{2}$

2. Reduce $\frac{36}{48}$ to its lowest terms.

$\frac{36}{48} = \frac{36 \div 2}{48 \div 2}$ reduce by 2

$= \frac{18}{24}$ reduce by 2

$= \frac{18 \div 2}{24 \div 2}$ reduce by 2

$= \frac{9}{12}$ reduce by 3

$= \frac{9 \div 3}{12 \div 3}$ reduce by 3

$= \frac{3}{4}$

Activity

Reduce the following fractions to their lowest terms.

1) $\frac{2}{4}$, 2) $\frac{3}{6}$ 3) $\frac{20}{30}$

4) $\frac{45}{90}$, 5) $\frac{25}{100}$ 6) $\frac{25}{100}$

Ordering Fractions

Example 1.

Arrange $\frac{1}{3}$, $\frac{1}{2}$ and $\frac{1}{4}$ in ascending order.

Look for the LCD of all the denominators.

LCD :- Lowest Common Divisor/Denominator.

$\frac{1}{3}$, $\frac{1}{2}$, $\frac{1}{4}$ lcd of 3, 2 and 4

M3 = 3, 6, 9, 12, 15, ...

M2 = 2, 4, 6, 8, 10, 12, 14, 16, ...

M2 = 4, 8, 12, 16, 20, ...

The LCD of the denominator is 12
Then Multiply it by every fraction.

$\frac{1}{\cancel{3}_1} \times \cancel{12}^4$ $\quad$ $\frac{1}{\cancel{2}_1} \times \cancel{12}^6$ $\quad$ $\frac{1}{\cancel{4}_1} \times \cancel{12}^3$

4 $\quad$ 6 $\quad$ 3

2 2 $\quad$ 3 3 $\quad$ 1 1

The order is $\frac{1}{4}$, $\frac{1}{3}$, $\frac{1}{2}$

Activity

1. Arrange the following fractions in descending order.

a) $\frac{3}{4}$, $\frac{2}{4}$, $\frac{1}{2}$ $\qquad$ b). $\frac{1}{2}$, $\frac{1}{4}$, $\frac{1}{9}$

2. Arrange the following fractions in ascending order.

a). $\frac{1}{5}$, $\frac{1}{2}$, $\frac{1}{6}$, $\frac{1}{3}$ $\qquad$ b) $\frac{7}{10}$, $\frac{3}{4}$, $\frac{3}{5}$

Adding fractions with different denominators.

Example 1.

Add: $\frac{2}{3} + \frac{1}{6}$ (Using equivalent fraction)

$\frac{2\times2}{3\times2} = \frac{4}{6}$, $\frac{2\times3}{3\times3} = \frac{6}{9}$, $\frac{2\times4}{3\times4} = \frac{8}{12}$

$\frac{1}{6} = \frac{1\times2}{6\times2} = \frac{2}{12}$, $\frac{1\times3}{6\times3} = \frac{3}{18}$, $\frac{1\times4}{6\times4} = \frac{5}{6}$

$\frac{2}{3} + \frac{1}{6} = \frac{8}{12} + \frac{2}{12} = \frac{8+2}{12} = \frac{10}{12} = \frac{5}{6}$

Using LCD

$\frac{2}{3} + \frac{1}{6}$ LCD $= 6$ $\quad \frac{(6\div3)\times2 + (6\div6)\times 1}{6} = \frac{2\times2 + 1\times 1}{6} = \frac{4+1}{6} = \frac{5}{6}$

Example 2.

Add: $\frac{1}{2} + \frac{1}{4}$ (use LCD)

$$\frac{(4\div2)\times2 + (4\div4)\times 1}{4}$$

$$= \frac{2\times1 + 1\times 1}{4} = \frac{2+1}{4} = \frac{3}{4}$$

Activity

Work out:

1. $\frac{2}{3} + \frac{1}{6}$

2. $\frac{5}{6} + \frac{3}{4}$

3. $\frac{1}{2} + \frac{1}{7}$

5. $\frac{3}{4} + \frac{2}{5}$

6. $\frac{2}{5} + \frac{3}{8}$

7. Find the sum of $\frac{5}{7}$ and $\frac{2}{3}$.

8. Simplify: $\frac{2}{5} + \frac{1}{4}$

Adding mixed fractions.

- Add the wholes and fractions separately or

Change the fractions to Improper fractions.

Examples.

1. $2\frac{3}{4} + 1\frac{1}{2}$ (By adding whole numbers)

$$2\frac{3}{4} + 1\frac{1}{2} = (2 + 1) + \frac{3}{4} + \frac{1}{2}$$
$$= 3 + \frac{3}{4} + \left(\frac{1 \times 2}{2 \times 2}\right)$$
$$= 3 + \frac{3}{4} + \frac{2}{4}$$
$$= 3 + \frac{5}{4}$$
$$= 3 + 1 + \frac{1}{4}$$
$$= 4\frac{1}{4}$$

Method II.

LCD = 4

1. $2\frac{3}{4} + 1\frac{1}{2}$

$$= \frac{11}{4} + \frac{3}{2}$$
$$= \frac{11}{4} + \frac{6}{4}$$
$$= \frac{17}{4} = 4\frac{1}{4}$$

Activity

Work out:

1. $1\frac{1}{2} + 2\frac{2}{5}$

2. $3\frac{2}{3} + 1\frac{1}{4}$

3. $2\frac{2}{5} + 1\frac{2}{3}$

4. $3\frac{1}{2} + 2\frac{2}{3}$

5. $1\frac{3}{4} + 4\frac{1}{7}$

6. $5\frac{3}{4} + 2\frac{1}{2}$

7. $3\frac{1}{4} + 1\frac{2}{5}$

8. $2\frac{1}{3} + 1\frac{1}{3}$

Subtracting fractions with different denominators.

Example 1.

denominator is 20 which is the lcm of 4 and 5

1. $\frac{3}{4} - \frac{2}{5} = \frac{(20\div4)\times3 - (20\div5)\times2}{20}$

$\frac{(5\times3) - (4\times2)}{20}$

$= \frac{15 - 8}{20}$

$= \frac{7}{20}$

Example 2.

2. $\frac{5}{6} - \frac{2}{3}$

$\frac{(6\div6)\times5 - (6\div3)\times2}{6}$

$= \frac{1\times5 - 2\times2}{6}$

$= \frac{5 - 4}{6} = \frac{1}{6}$

Activity

Work out:

1. $\frac{3}{5} - \frac{3}{9}$

2. $\frac{2}{3} - \frac{1}{5}$

3. $\frac{1}{2} - \frac{1}{3}$

4. $\frac{1}{2} - \frac{1}{4}$

5. $\frac{5}{6} - \frac{1}{10}$

6. $\frac{4}{3} - \frac{1}{4}$

7. $\frac{2}{3} - \frac{1}{2}$

Subtracting mixed fractions.

Example 1.

1. $3\frac{1}{2} - 1\frac{2}{3}$

$= 3 - 1 + \left(\frac{1}{2} + \frac{1}{3}\right)$

$= 2 + \frac{1}{6} \quad \frac{12 + 1}{6}$

$\frac{13}{6} = 2\frac{1}{6}$

Method II

$3\frac{1}{2} - 1\frac{2}{3} = \frac{7}{2} - \frac{4}{3}$ LCD is 6

$\frac{(6 \div 2)x7 - (6 \div 3)x\,4}{6}$

$= \frac{3x7 - 2x4}{6}$

$= \frac{21 - 8}{6} = \frac{13}{6} = 2\frac{1}{6}$

2.

$5\frac{1}{4} - 2\frac{1}{5} = \frac{21}{4} - \frac{11}{5}$ LCD is 6

$\frac{(20 \div 4)x2 - (20 \div 5)x11}{20}$

$\frac{5 \times 21 - 11 \times 4}{20}$

$\frac{105 - 44}{20} = \frac{61}{20} = 2\frac{3}{4}$

Activity

Work out the following.

1. $3\frac{1}{3} - 1\frac{2}{3}$

2. $2\frac{1}{2} - 1\frac{3}{10}$

3. $2\frac{1}{2} - \frac{3}{7}$

4. $5\frac{3}{4} - 2\frac{1}{3}$

5. $3\frac{3}{4} - 2\frac{1}{3}$

6. $6\frac{5}{8} - 2\frac{2}{7}$

7. $5 - \frac{2}{7}$

8. $3\frac{2}{5} - 1\frac{3}{5}$